

Chapter 1: Stereochemistry

- **Unit description:**

- This unit deals with 3D aspects of molecules including (stereo)isomerism, chirality, optical activity, enantiomers and diastereomers and meso compounds.

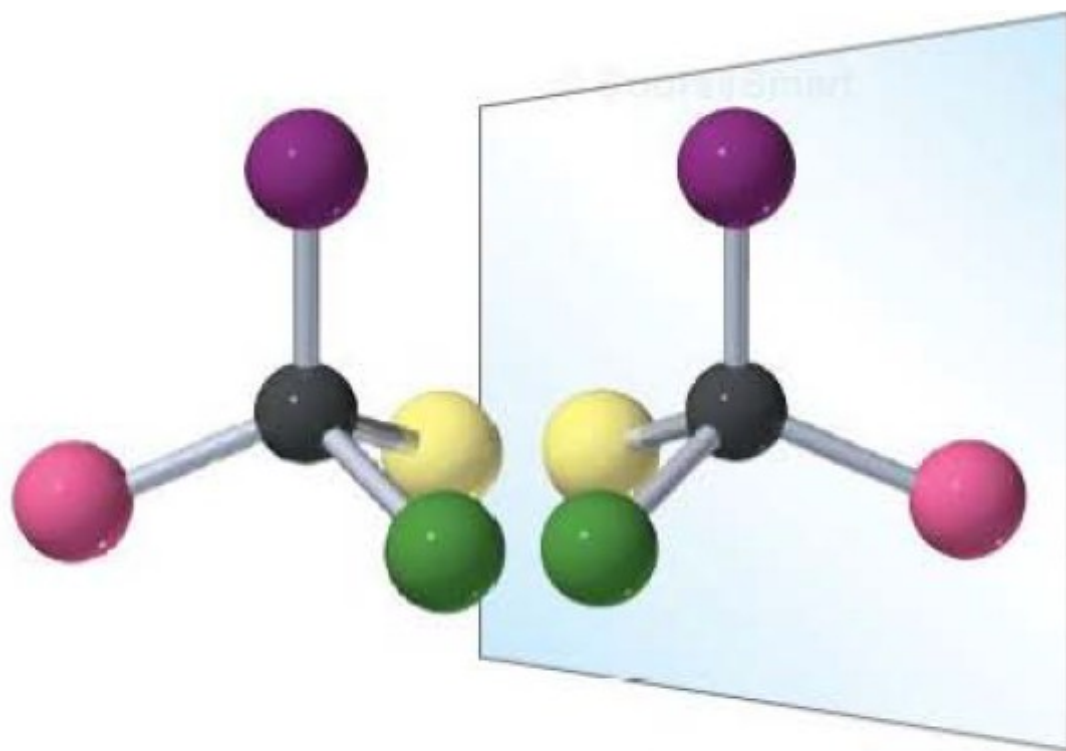
- **Unit learning outcomes:**
 - Define stereochemistry
 - Explain isomerism in simple organic molecules
 - Explain the role of chirality in stereoisomerism
 - Describe enantiomers
 - Apply the R-S system in naming enantiomers
 - Outline the use of polarimeter (optical activity) in differentiating enantiomers
 - Outline resolution of enantiomers
 - Describe diastereomers
 - Describe meso compounds
 - Explain biological significance of stereoisomerism

- **Contents:**

- 1.1. Introduction (Definition)
- 1.2. Isomerism (constitutional isomers, stereoisomerism, conformational and configurational isomers)
- 1.3. Chirality (Definition and methods to identify chiral center/molecules)
- 1.4. Enantiomers (Definition)
- 1.5. Naming enantiomers: The Cahn–Ingold–Prelog R–S notational system enantiomers,
- 1.6. Optical activity [observed and specific rotation])
- 1.7. Resolution of enantiomers
- 1.8. Diastereomers
- 1.9. Meso compounds
- 1.10. Biological significance of stereoisomerism

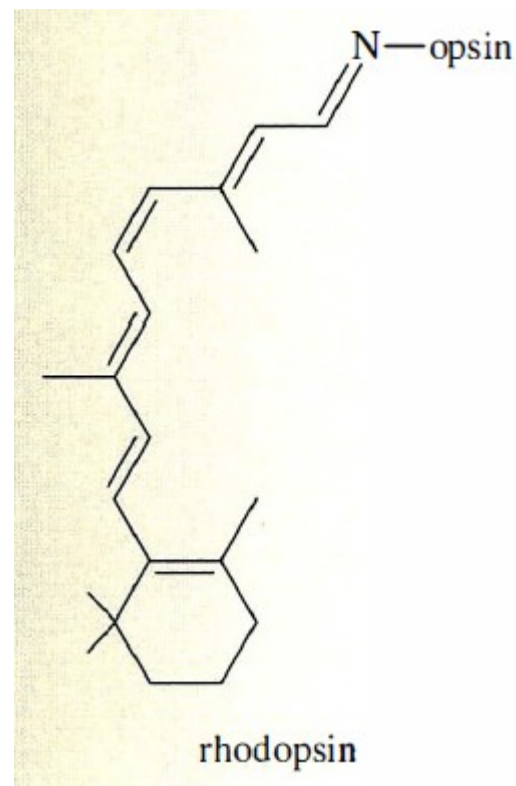
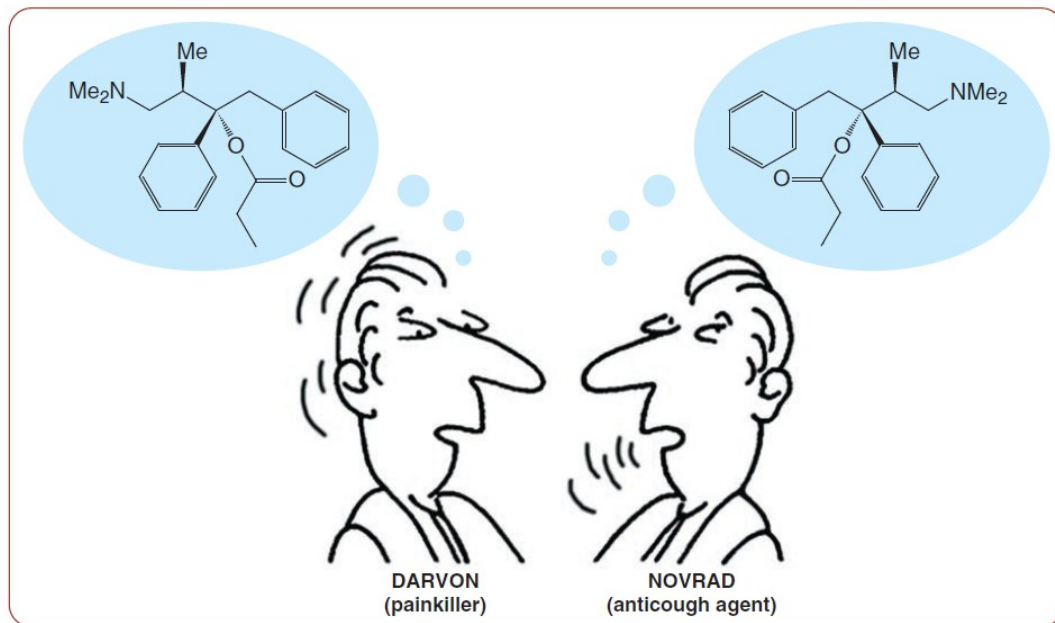
1.1. Introduction

- **Stereochemistry**
 - 3D chemistry

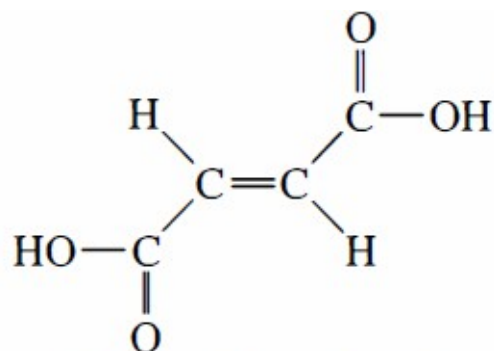


non-superimposable
mirror images

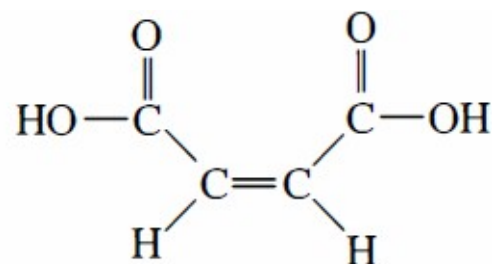
- **Biological action vs structure**



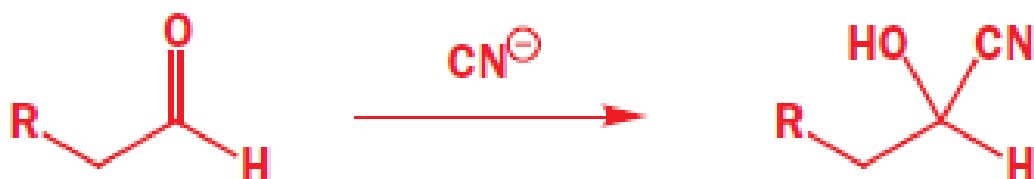
- **Physicochemical properties vs structure**



fumaric acid, mp 287°C
essential metabolite

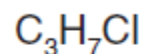


maleic acid, mp 138°C
toxic irritant

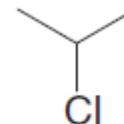


1.2. Isomerism

- **Isomers**
 - “Made of the same parts”

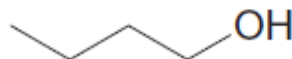
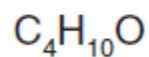


and

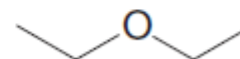


1-Chloropropane

2-Chloropropane

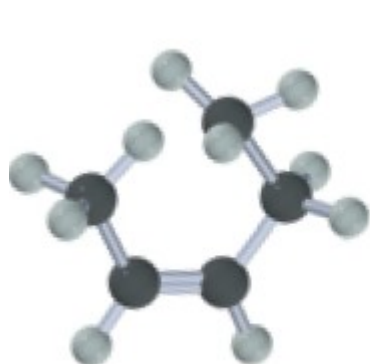


and

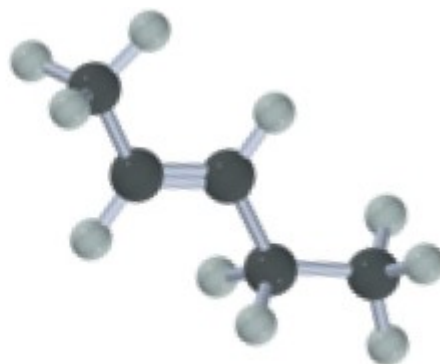


1-Butanol

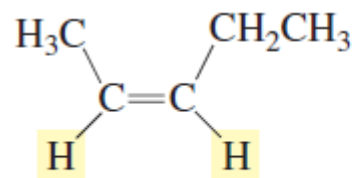
Diethyl ether



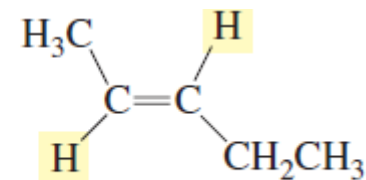
cis-2-pentene



trans-2-pentene



cis-2-pentene



trans-2-pentene

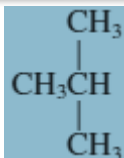
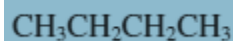
- Classes of isomers

Isomers

Have the same molecular formula, but different structures

Constitutional isomers

Differ in the order of attachment of atoms (Connectivity)

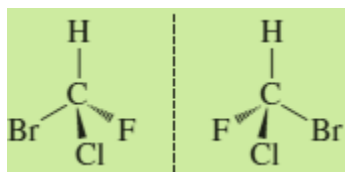


Stereoisomers

Atoms are connected in the same order but differ in spatial orientation

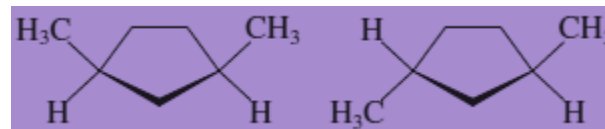
Enantiomers

Image and mirror image are not superposable



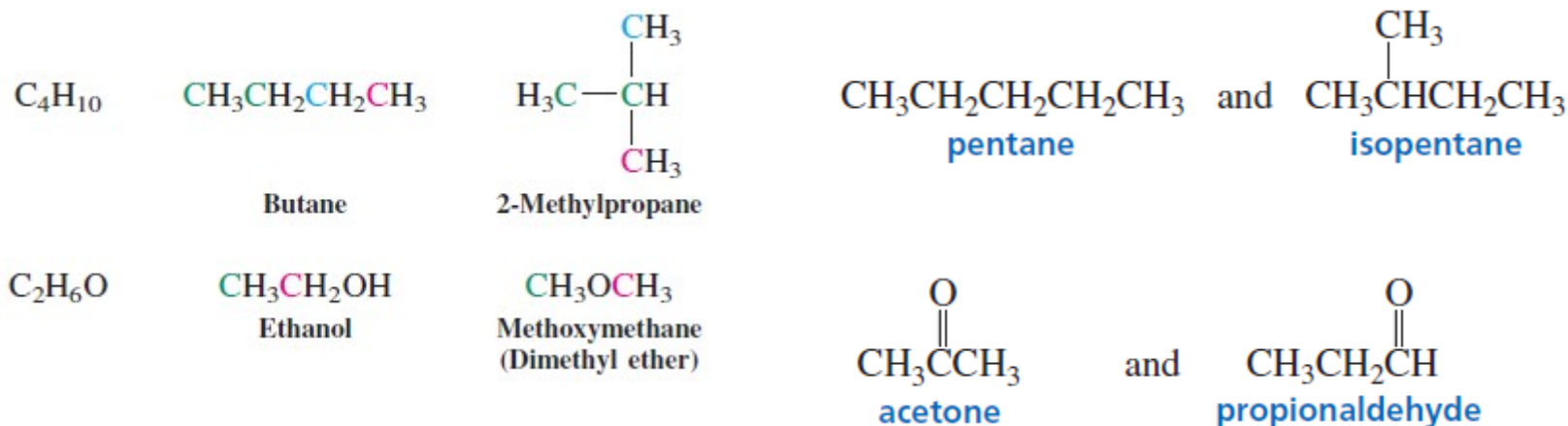
Diastereomers

Not related as image and mirror image



1.2.1. Constitutional isomers

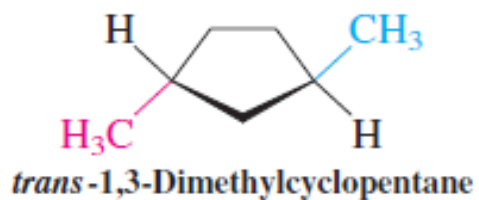
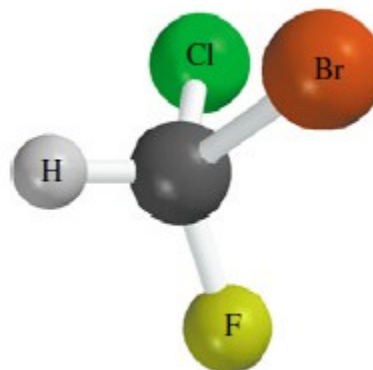
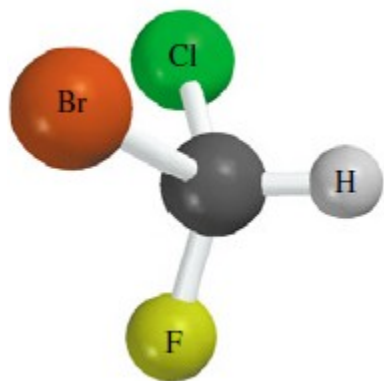
- Sequence (connectivity) difference



- Different compounds have entirely different physicochemical properties

1.2.2. Stereoisomerism

- Same connectivity but different 3D structure



- **Two types of stereoisomers**
 - Conformational isomers and
 - Configurational isomers

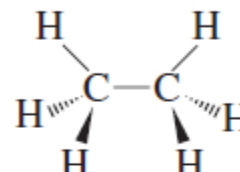
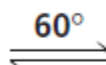
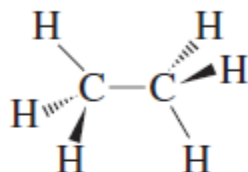
- **Conformational isomers**

- Example: Ethane

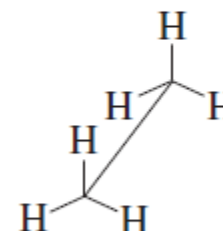
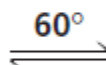
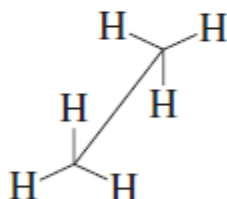
staggered conformation for rotation about the carbon-carbon bond in ethane

eclipsed conformation for rotation about the carbon-carbon bond in ethane

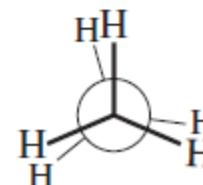
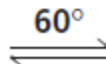
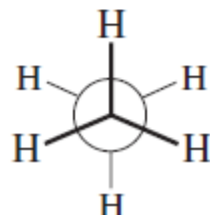
perspective formulas



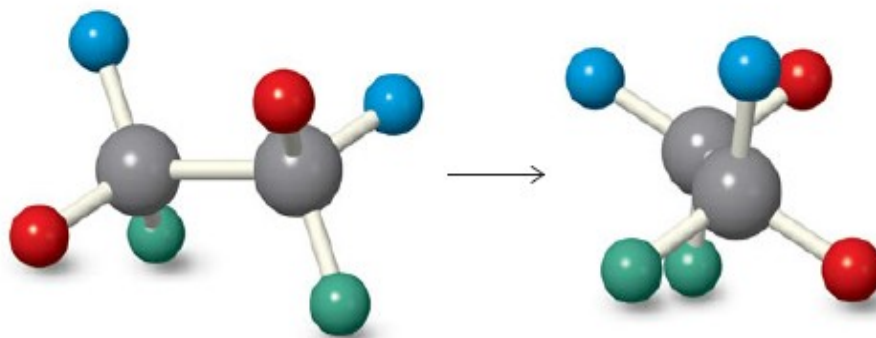
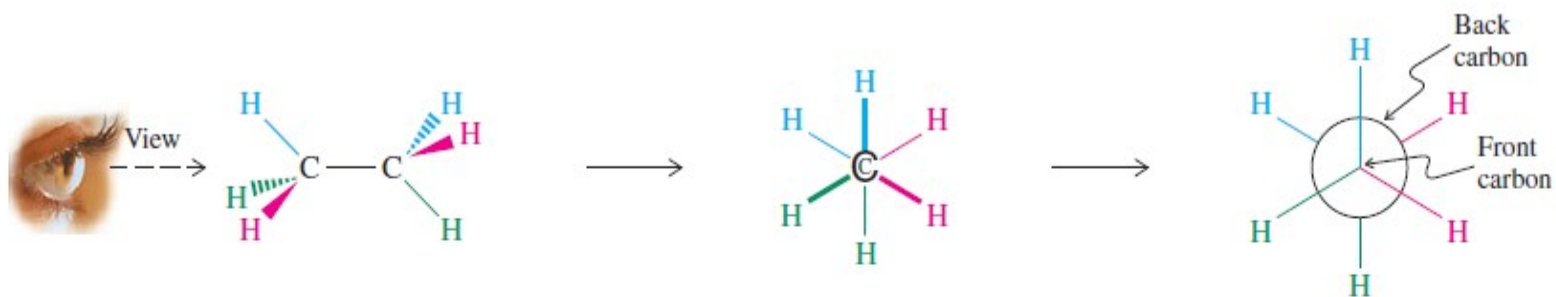
sawhorse projections



Newman projections

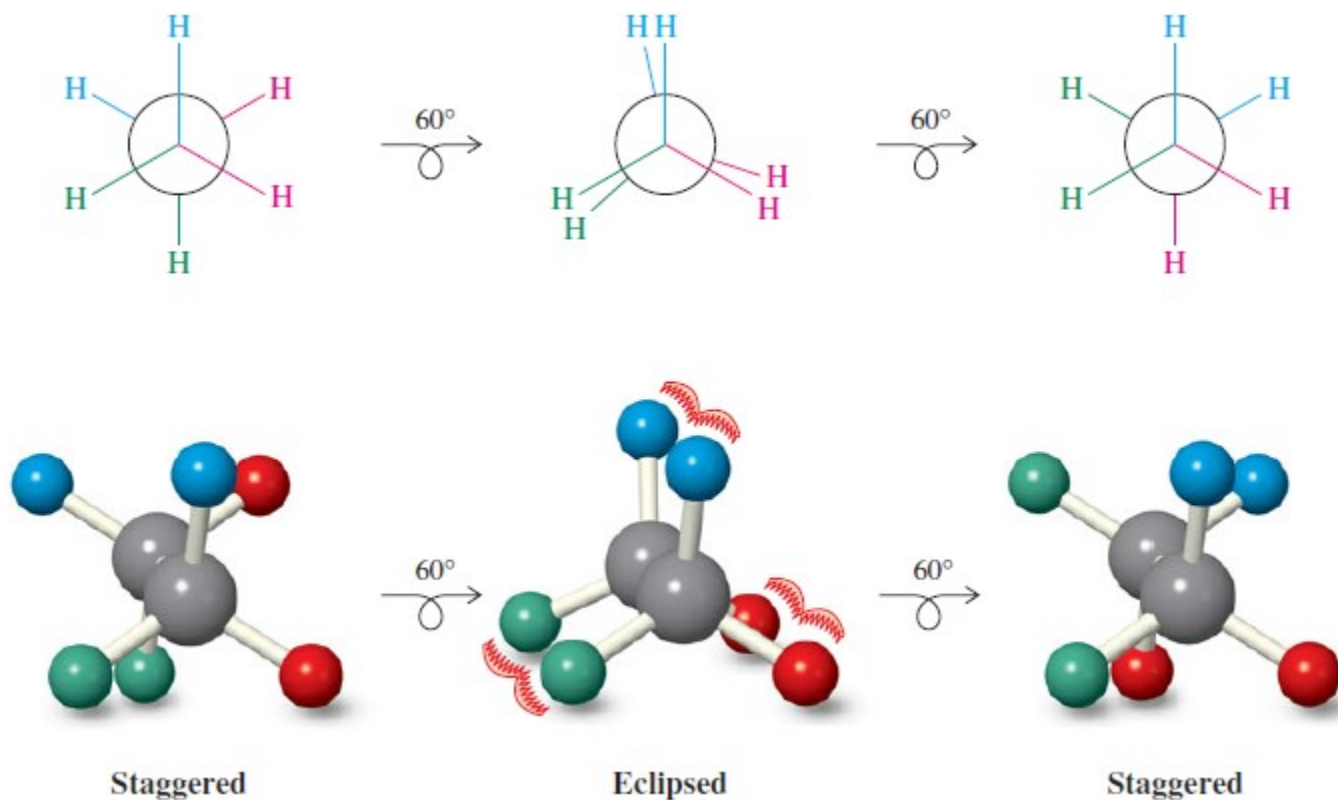


– Newman projections



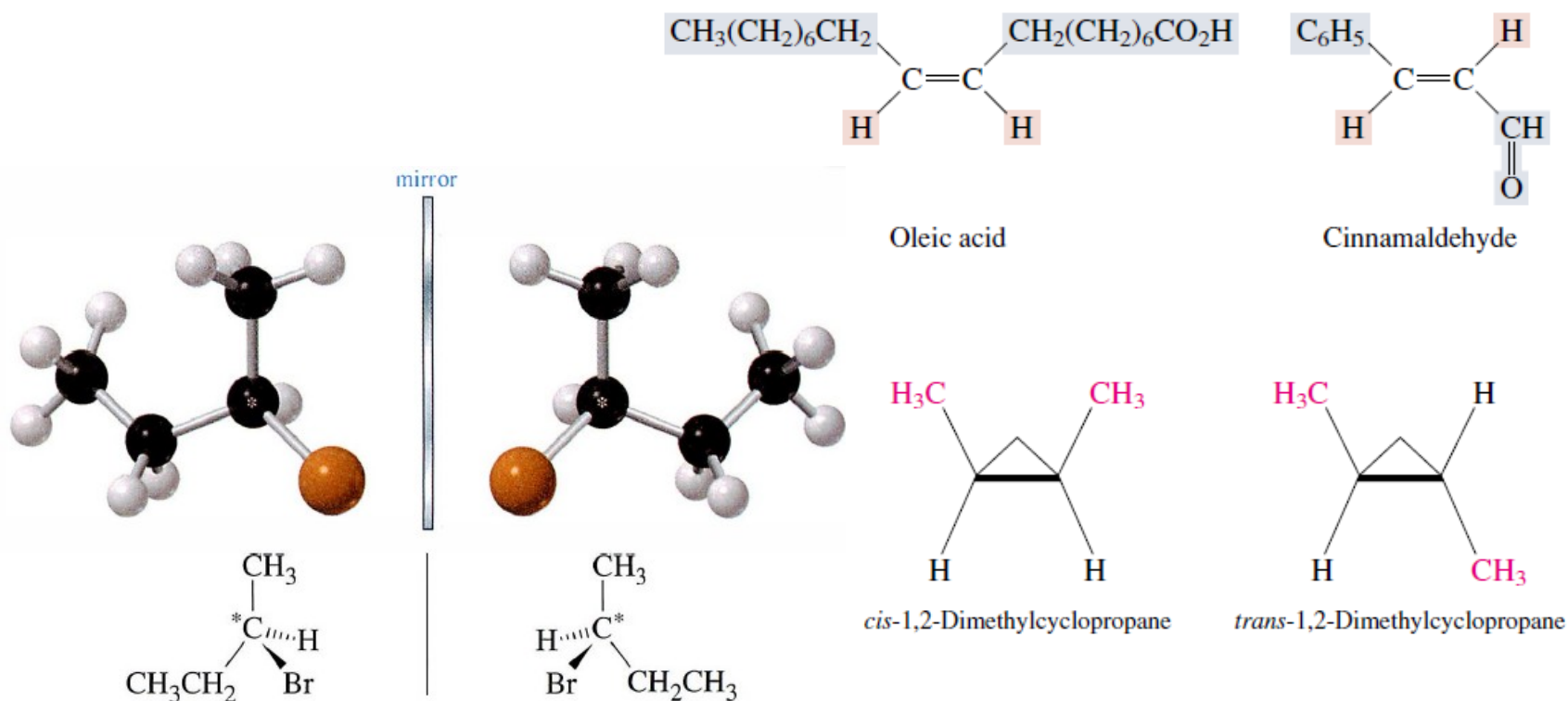
– Rotation around single bond

- Staggered vs Eclipsed



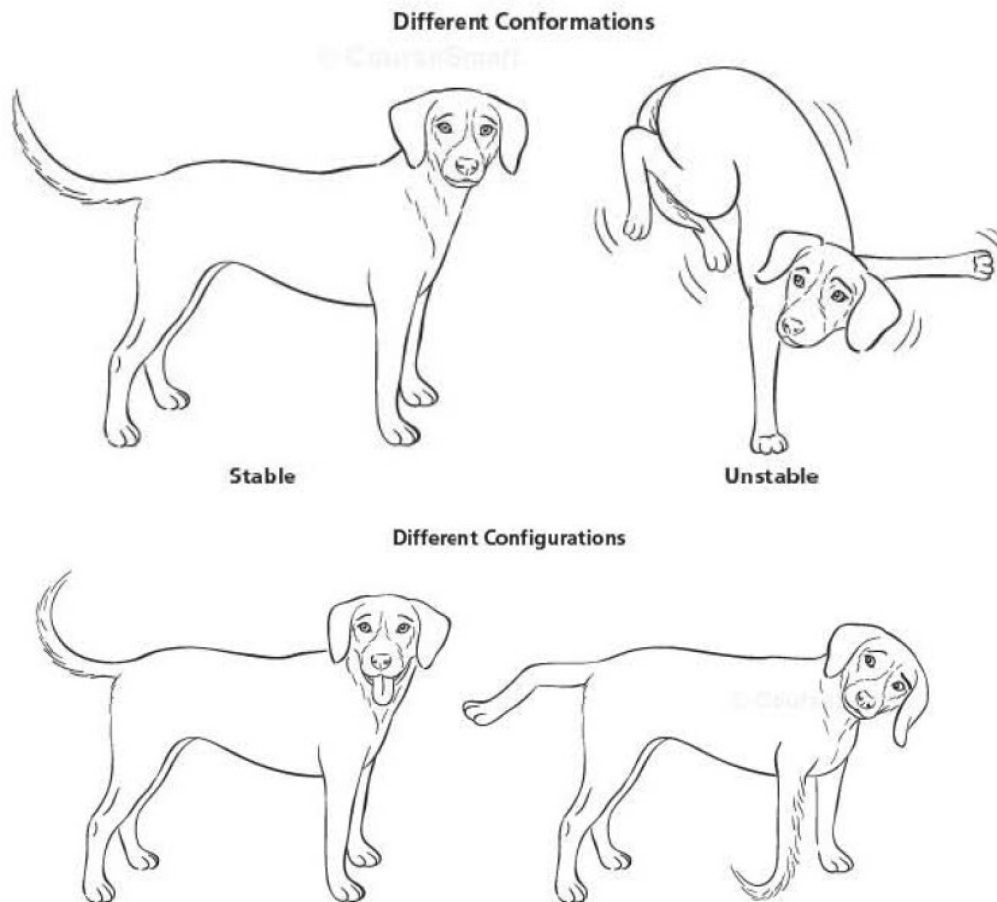
• Configurational (stereochemical) isomers

- Difference in spatial arrangement
- Cannot be converted from one into another by rotations about single bonds



Summary

- **Conformational vs configurational isomers**



1.3. Chirality

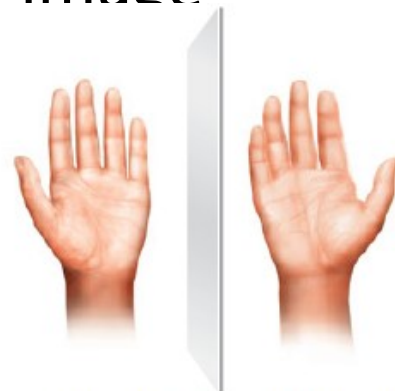
1.3. Chirality ...

- **Chirality**

- The property of *nonsuperimposability* of an object on its mirror image



The mirror image of a right hand is a left hand.



Left hand
Image

Mirror

Right hand
Mirror image



Nonsuperimposable
left and right hands



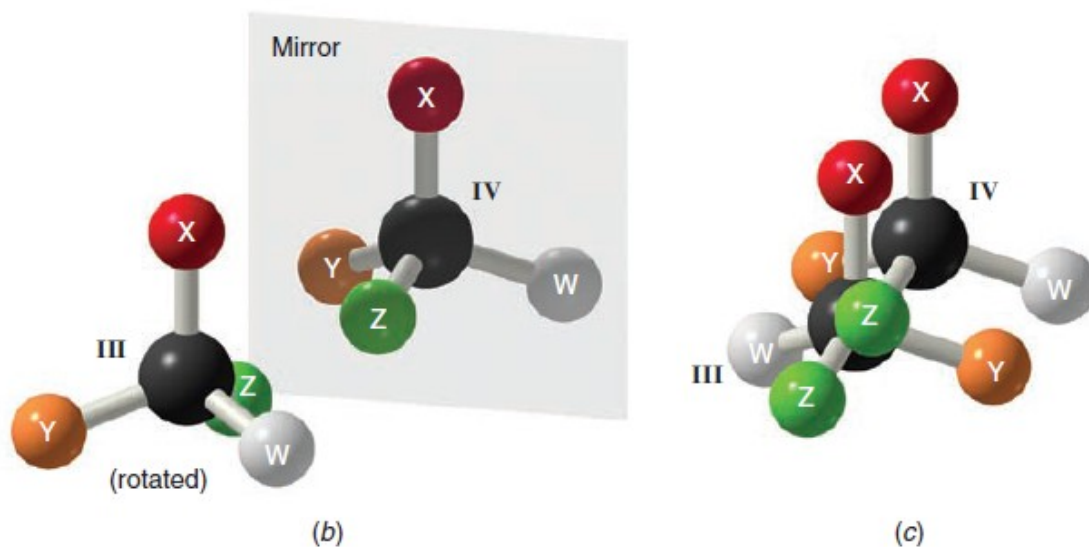
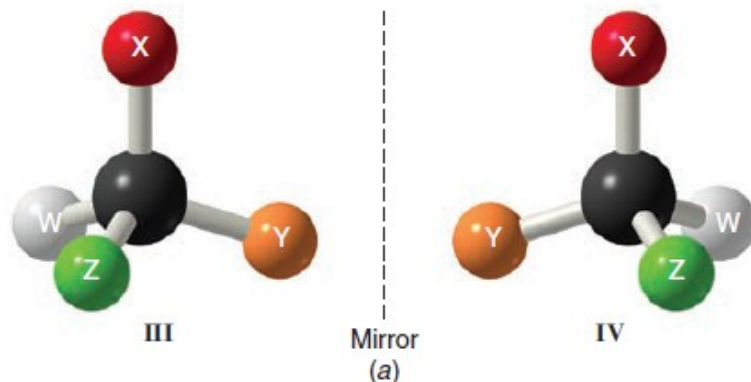
Mirror



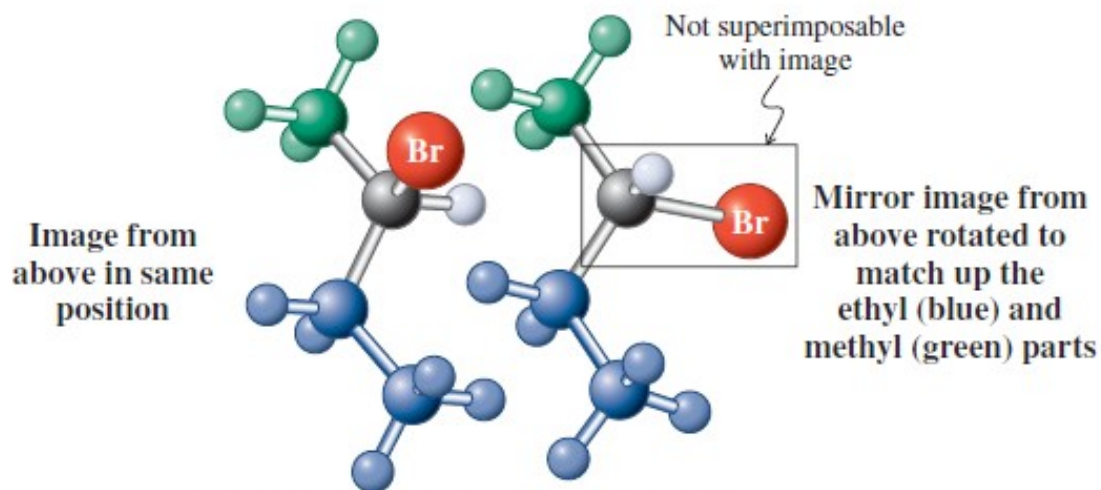
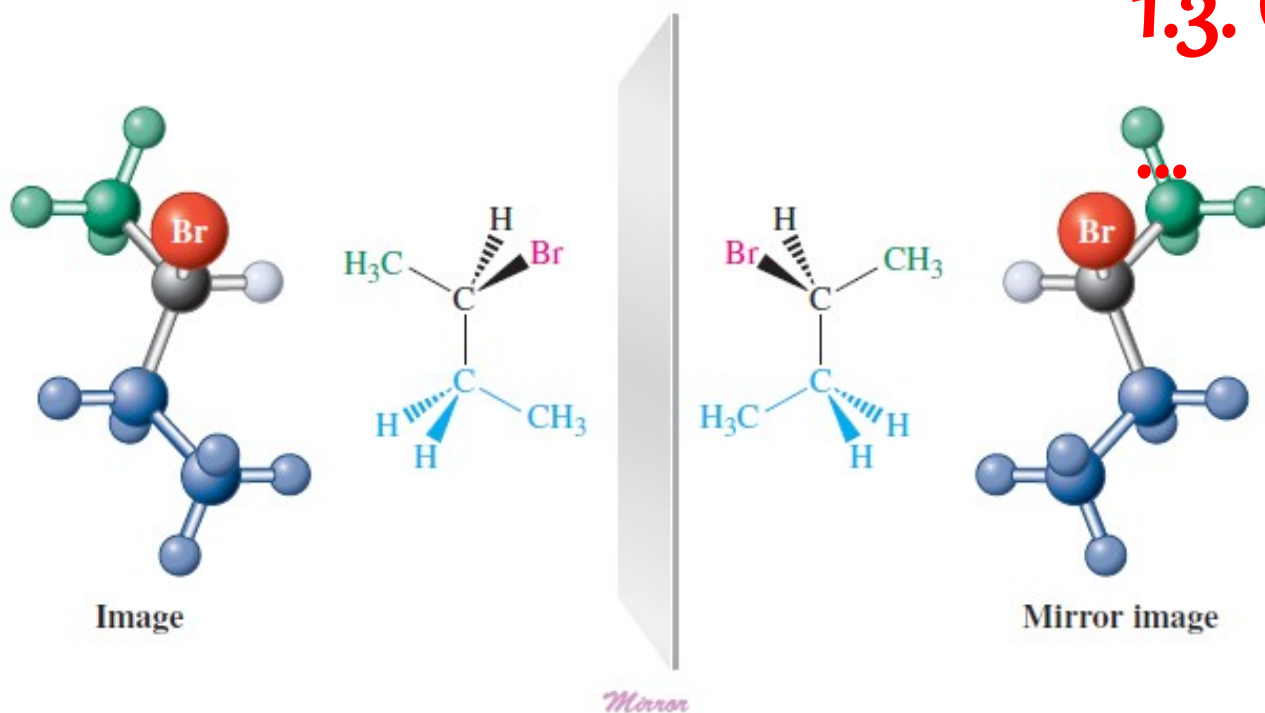
Superimposable image and
mirror image of hammer

1.3. Chirality

- Chiral: a molecule is not superimposable on its mirror image



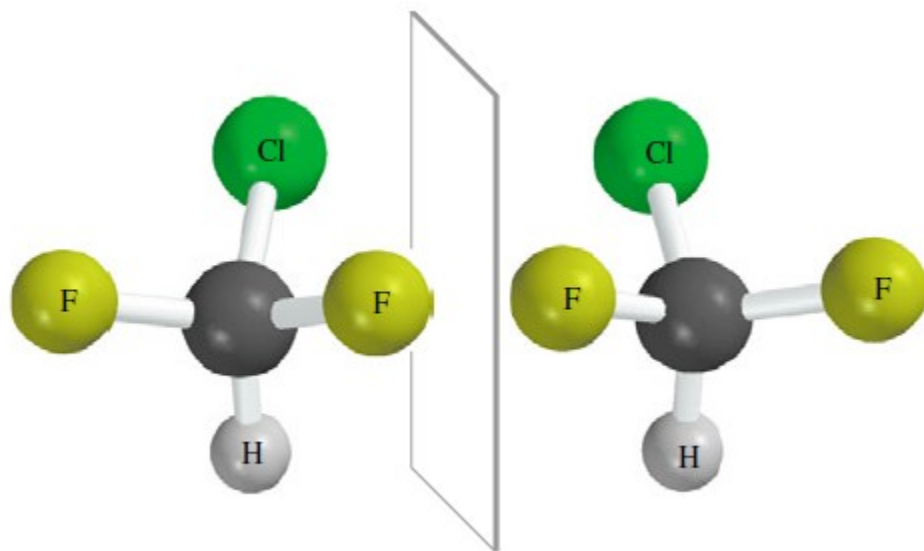
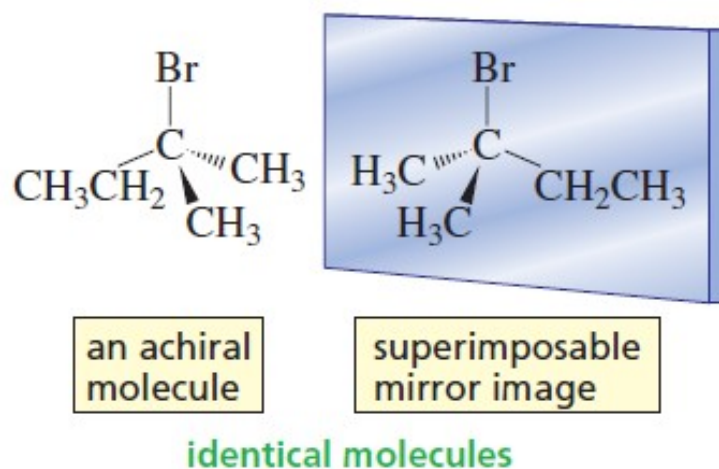
1.3. Chirality



The two enantiomers of 2-bromobutane are nonsuperimposable

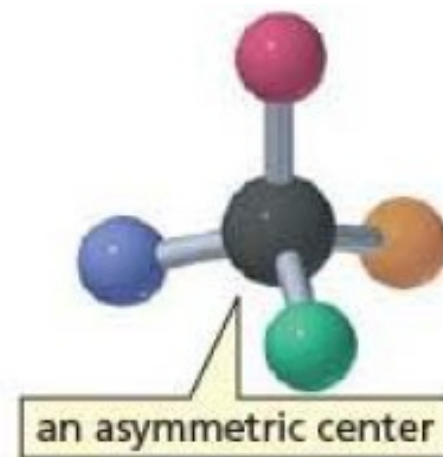
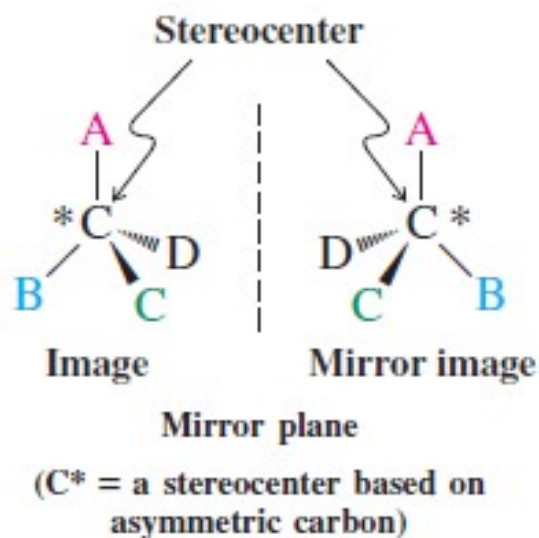
1.3. Chirality ...

- Achiral: superimposable on its mirror image**



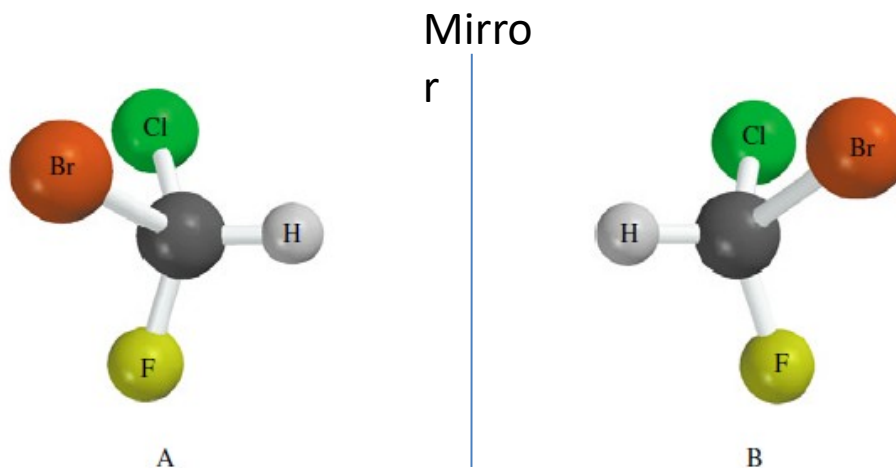
1.3. Chirality ...

- **Stereocenters, chirality center, or asymmetric carbons**
 - The most common cause for chirality

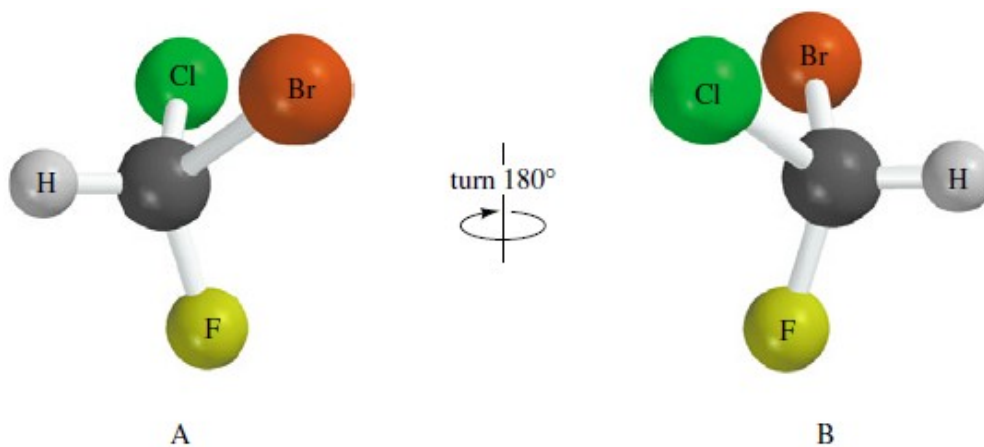


1.3. Chirality ...

– Example 1: bromochlorofluoromethane (BrClFCH)

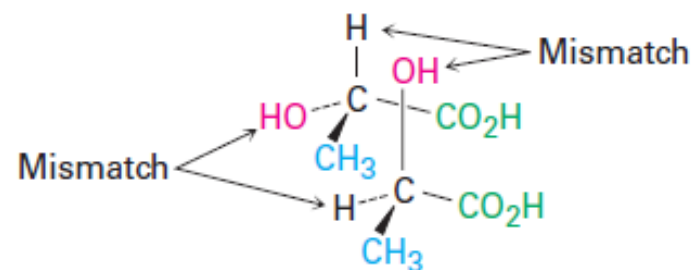
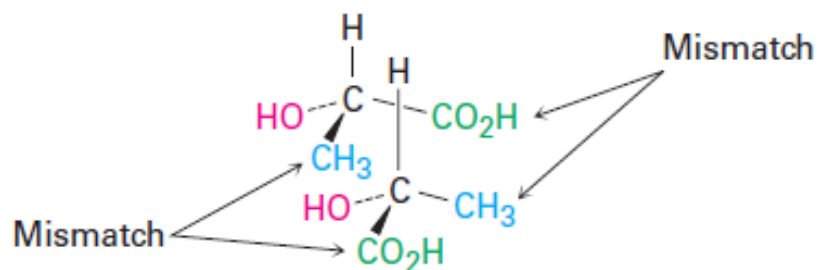
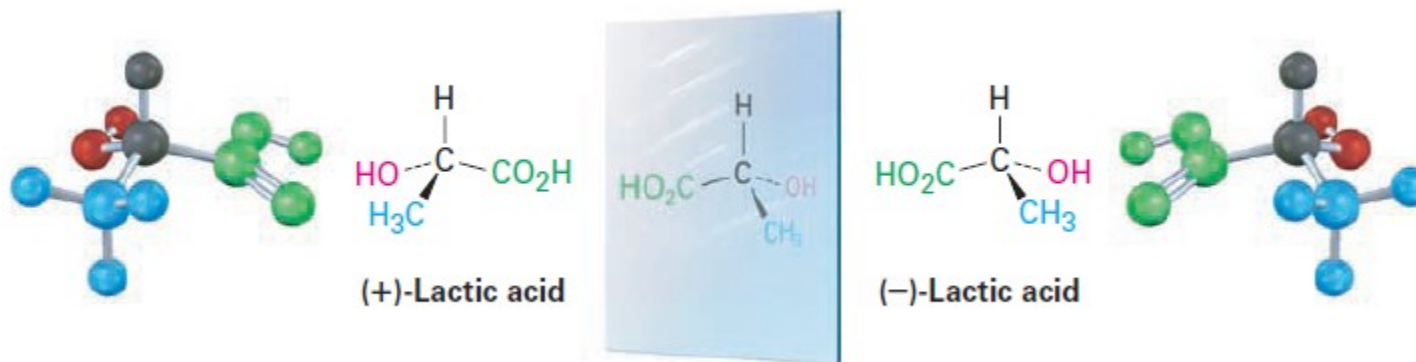


To test for superposability, reorient B by turning it 180°.



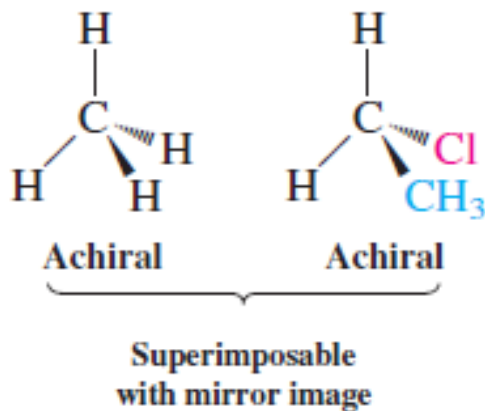
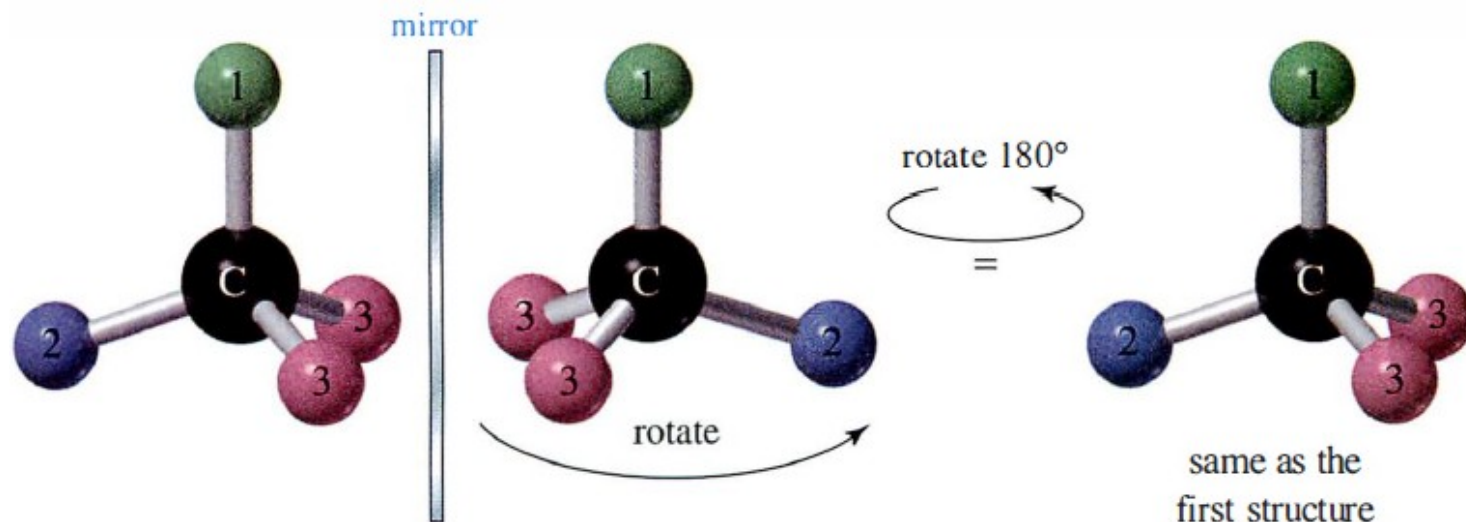
1.3. Chirality ...

– Example 2: Lactic acid



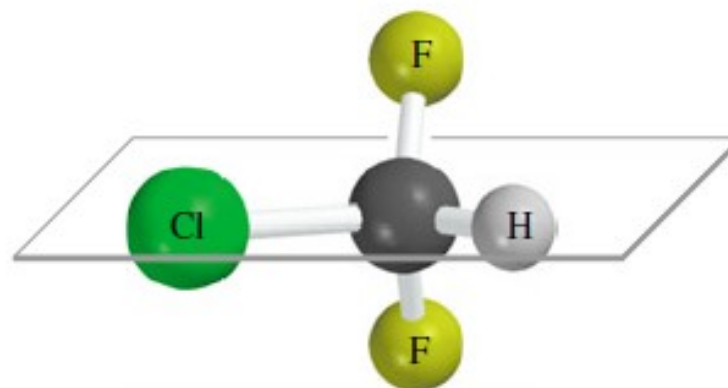
1.3. Chirality ...

– Example 3: achiral molecule



1.3. Chirality ...

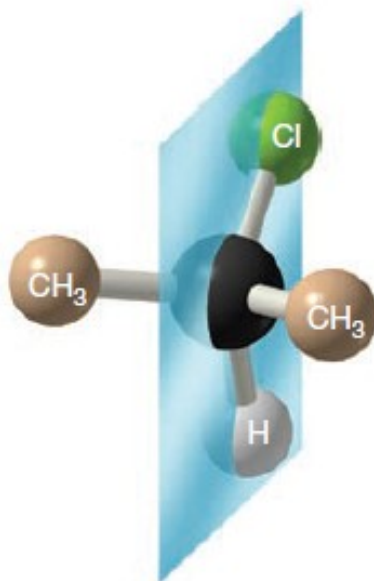
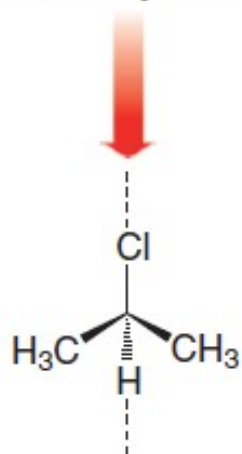
- **Test for chirality**
 - Plane of symmetry



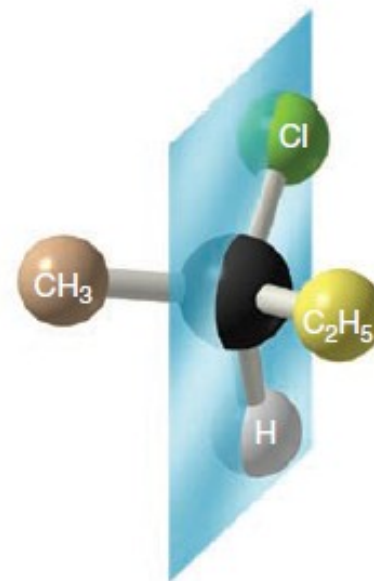
1.3. Chirality ...

– Example 1:

Plane of symmetry



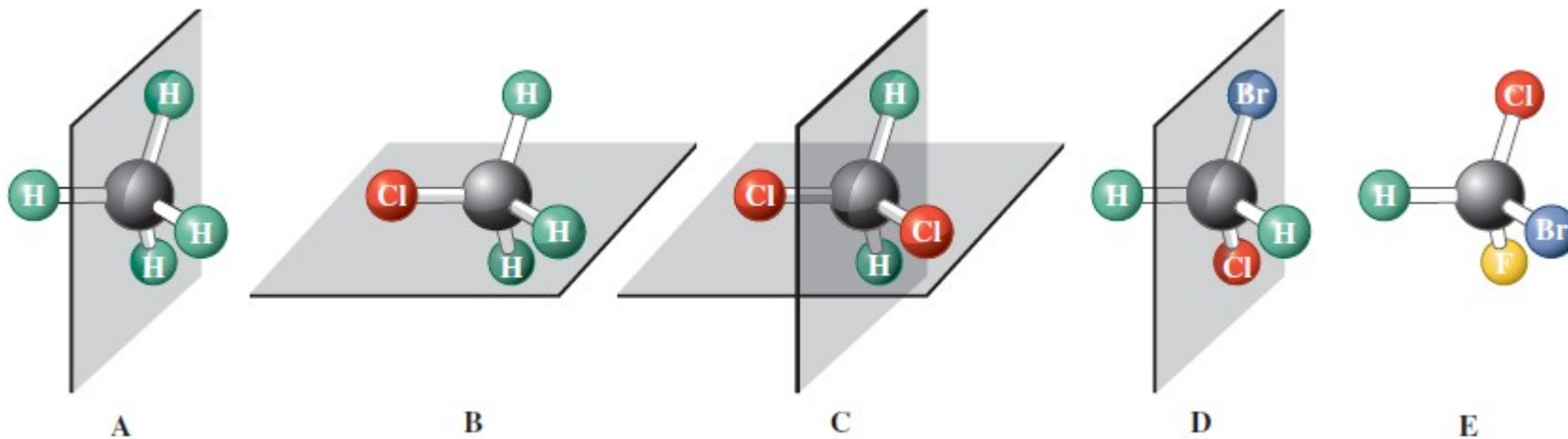
Achiral



Chiral

1.3. Chirality ...

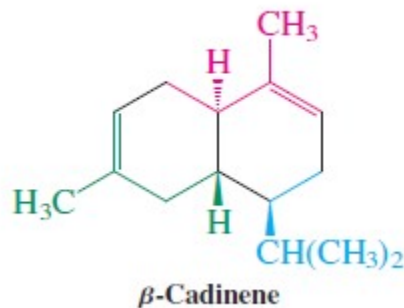
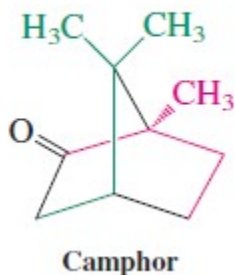
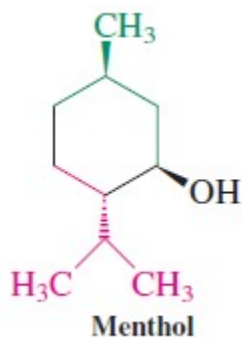
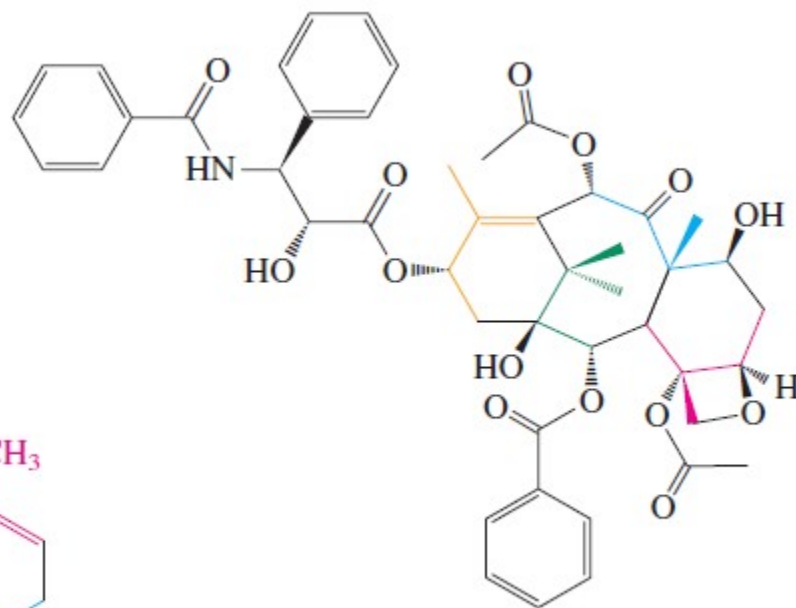
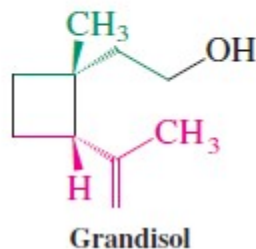
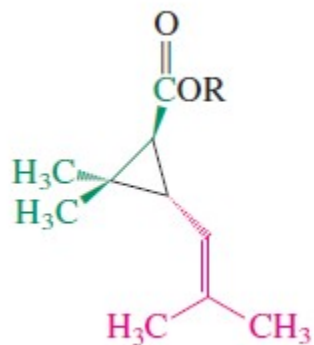
– Example 2:



1.3. Chirality ...

- **Exercise:**

- Among the natural products shown below, which are chiral and which are achiral?
- Give the number of stereocenters in each case.



Taxol

1.4. Enantiomers

- Classes of isomers

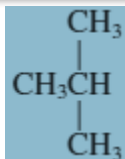
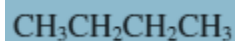
1.4. Enantiomers ...

Isomers

Have the same molecular formula, but different structures

Constitutional isomers

Differ in the order of attachment of atoms (Connectivity)

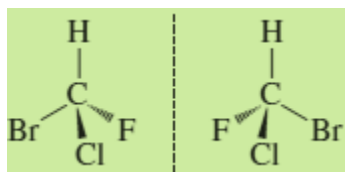


Stereoisomers

Atoms are connected in the same order but differ in spatial orientation

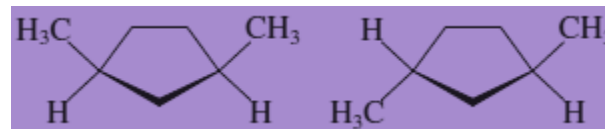
Enantiomers

Image and mirror image are not superposable



Diastereomers

Not related as image and mirror image

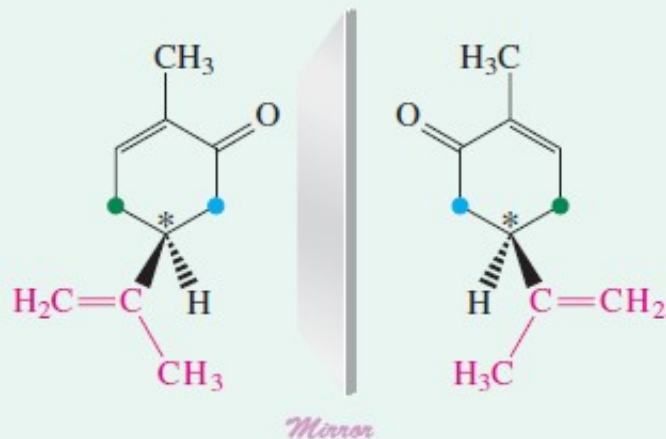


1.4. Enantiomers ...

- Enantiomers
 - Molecules that are non-superimposable mirror images of each other



Caraway seeds



Carvone enantiomers



Spearmint

1.4. Enantiomers ...

- *If a molecule is nonsuperimposable on its mirror image, the mirror image must be a different molecule.*
 - Therefore, *differ in structure only in the left and right handedness of their orientations.*
- **Have same physicochemical properties, except:**
 1. Optical activity
 2. Interaction with chiral molecules

1.4. Enantiomers ...

- **Pharmacological properties**
 1. Only one enantiomer is active
 2. Both enantiomers show essentially identical activities:
 3. Both enantiomers have similar activity, but one enantiomer is more active;
 4. Enantiomers show different pharmacological activities

1.5. Naming enantiomers:

The Cahn–Ingold–Prelog R–S notational system

1.5. Naming enantiomers...

- Developed by R. S. Cahn, C. Ingold, and V. Prelog



Vladimir Prelog

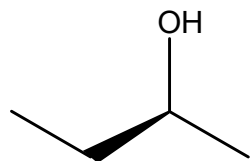


Christopher Kelk Ingold

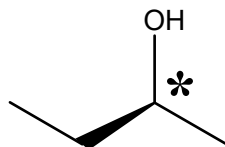
1.5. Naming enantiomers...

- Steps:**

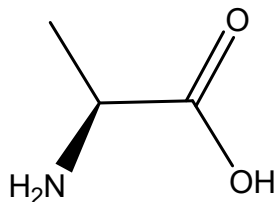
1. Locate the chiral center and identify the substituents on the chiral center



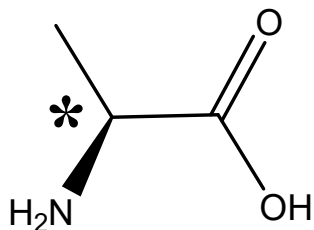
2-butanol



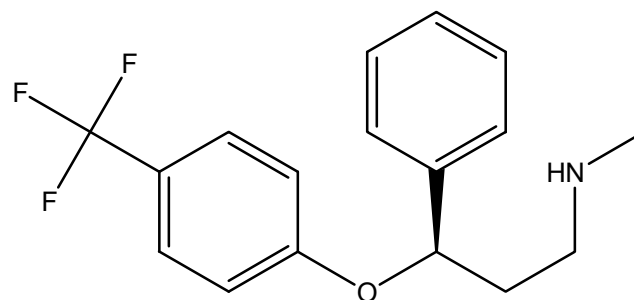
2-butanol



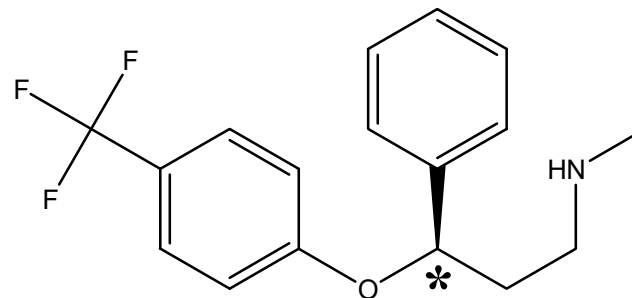
alanine



alanine



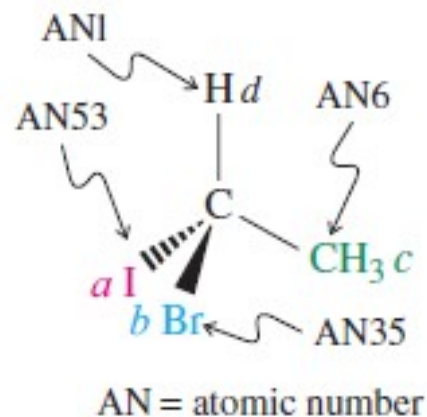
Fluoxetine



Fluoxetine

1.5. Naming enantiomers...

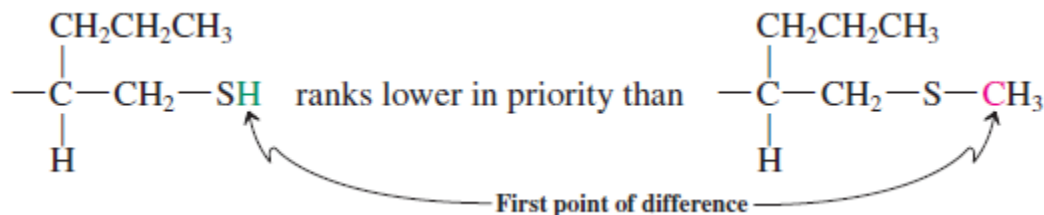
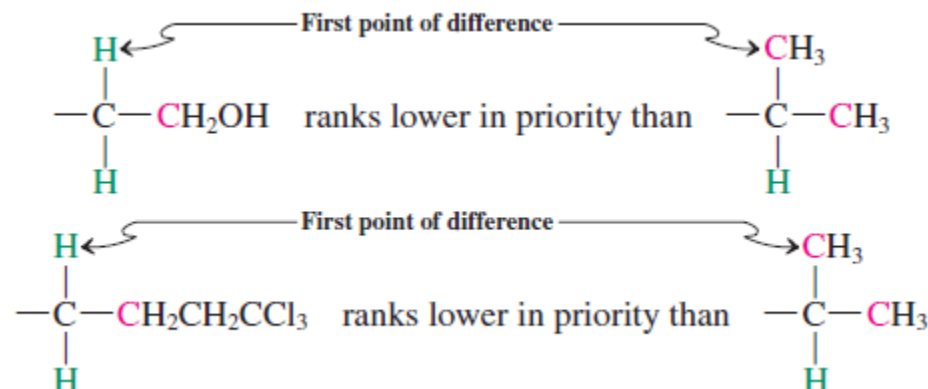
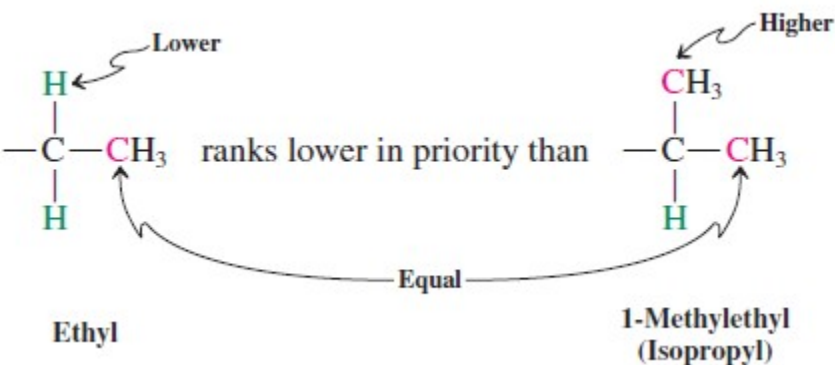
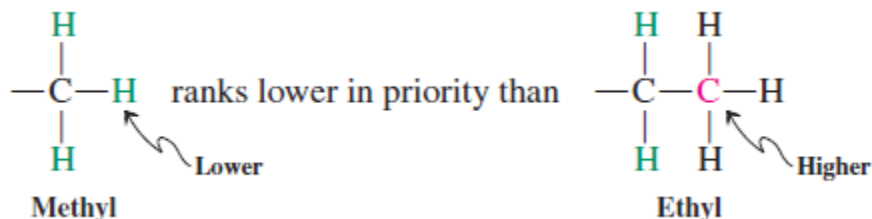
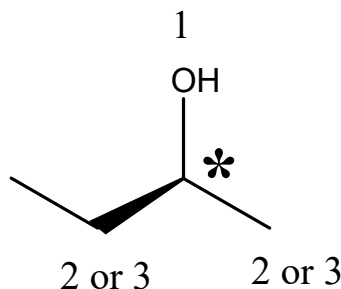
2. Assign a priority (i.e. 1, 2, 3 or 4) or preference (i.e. a, b, c, or d)



1.5. Naming enantiomers...

– Groups tied?

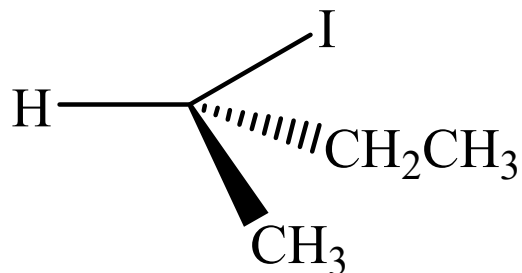
- Example:



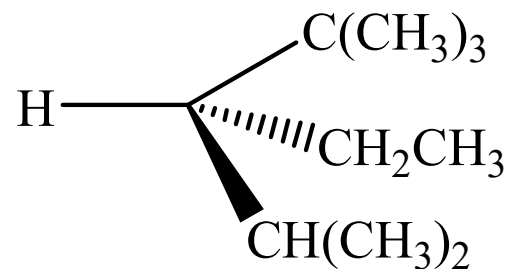
1.5. Naming enantiomers...

- Example:

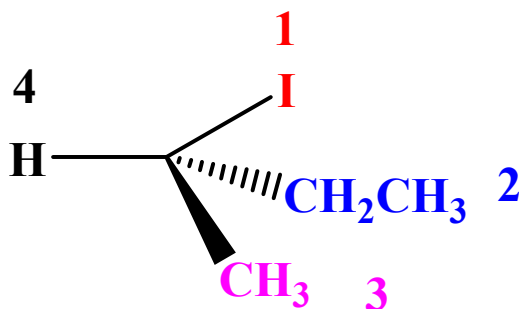
– 2-Iodobutane and 3-Ethyl-2,2,4-trimethylpentane



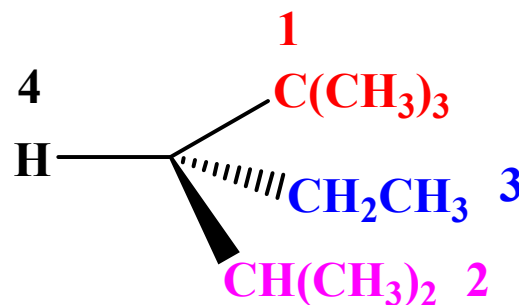
2-Iodobutane



3-Ethyl-2,2,4-trimethylpentane



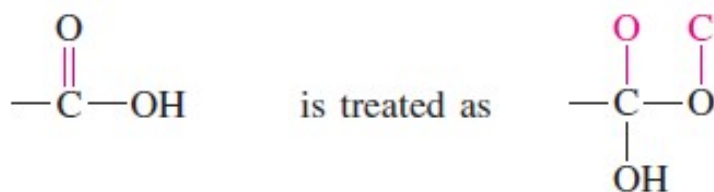
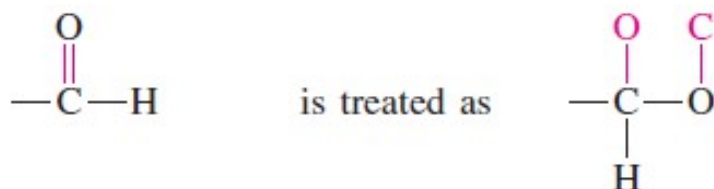
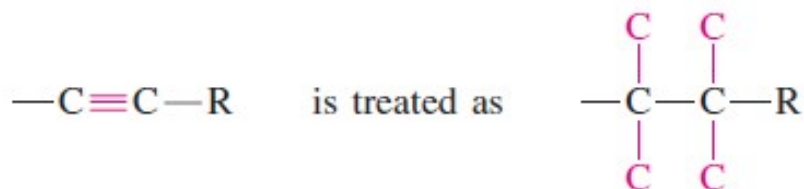
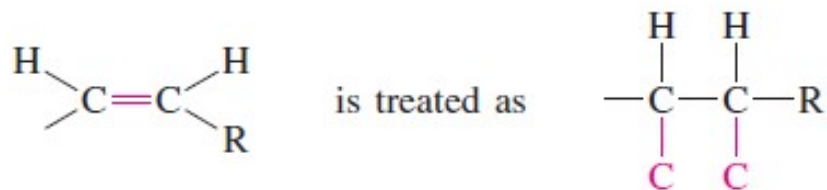
2-Iodobutane



3-Ethyl-2,2,4-trimethylpentane

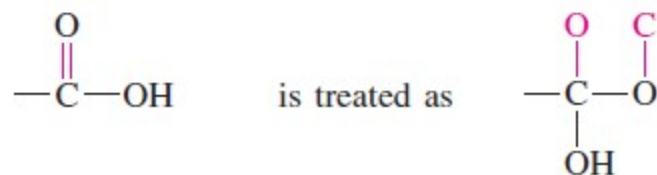
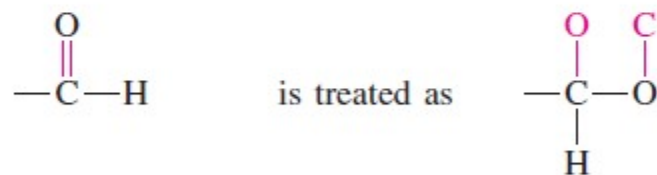
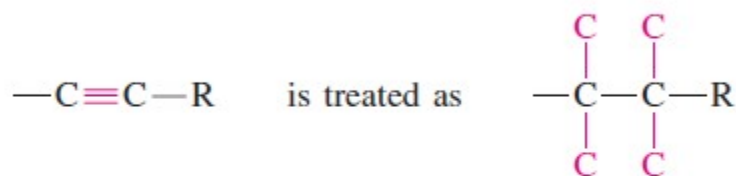
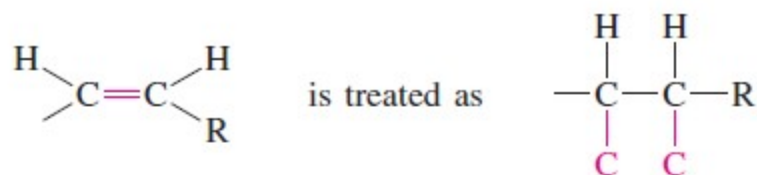
1.5. Naming enantiomers...

– Double and triple bonds

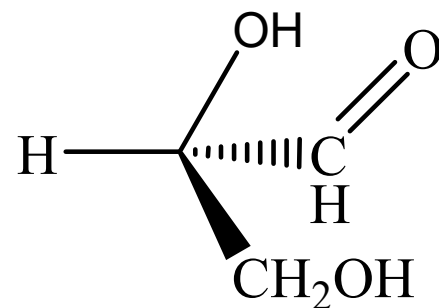
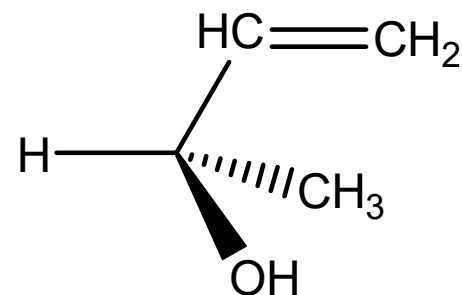


1.5. Naming enantiomers...

- Double and triple bonds

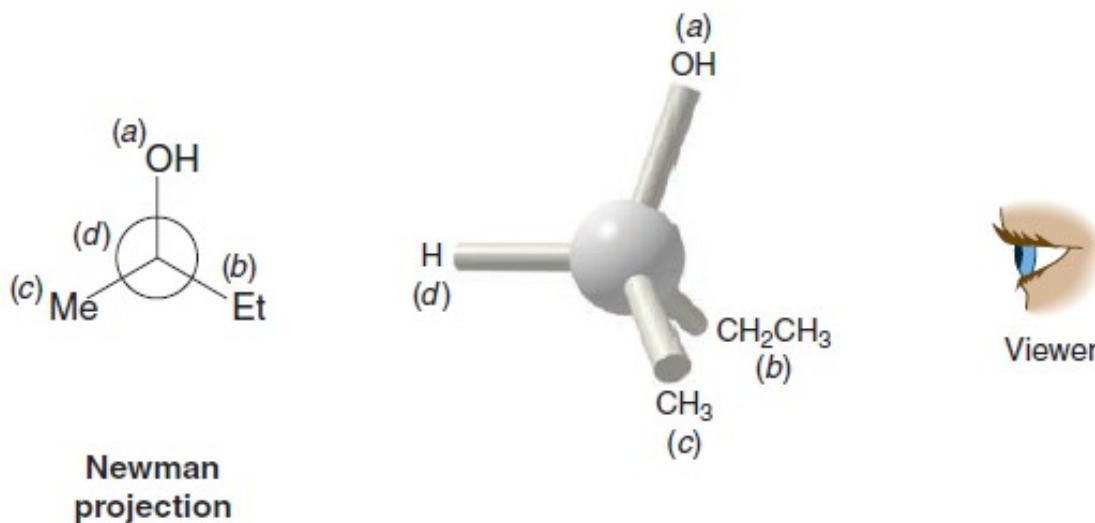


- Exercise: Give priority



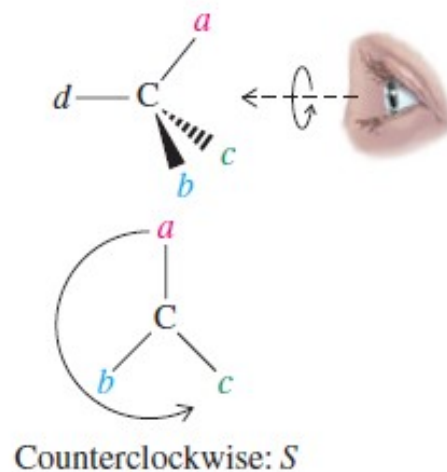
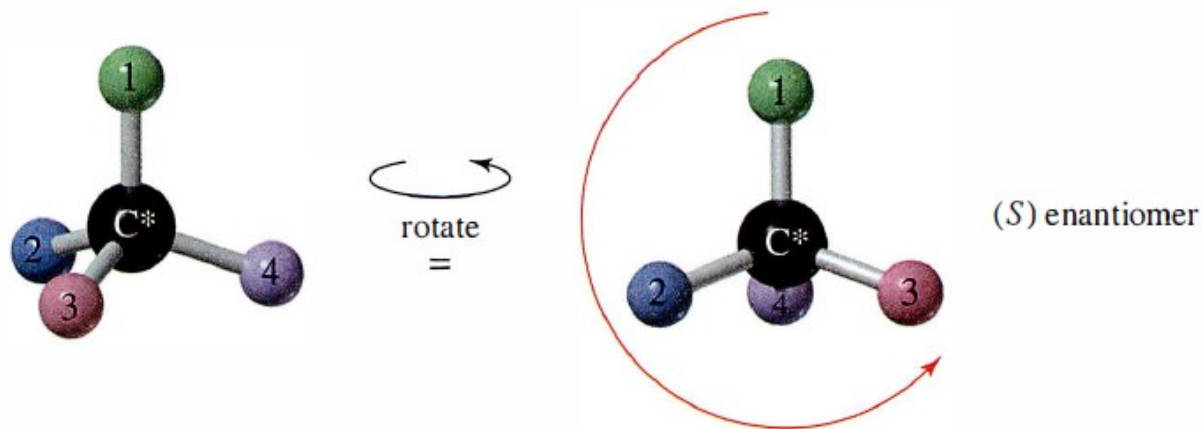
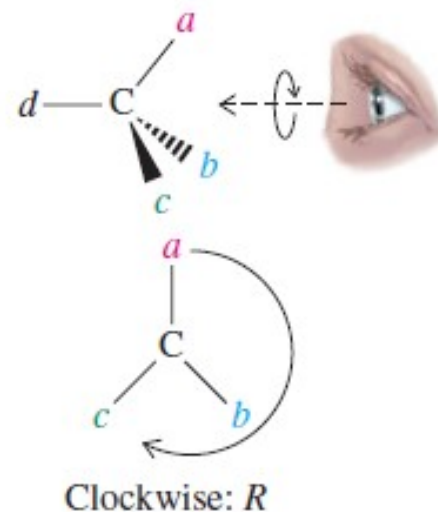
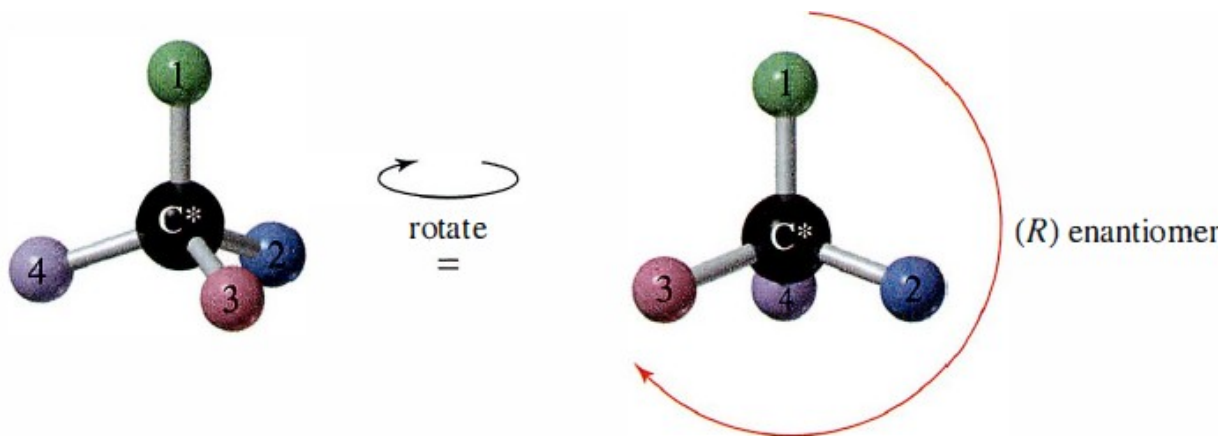
1.5. Naming enantiomers...

3. Arrange the molecule so that the lowest priority substituent is pointing away from you.

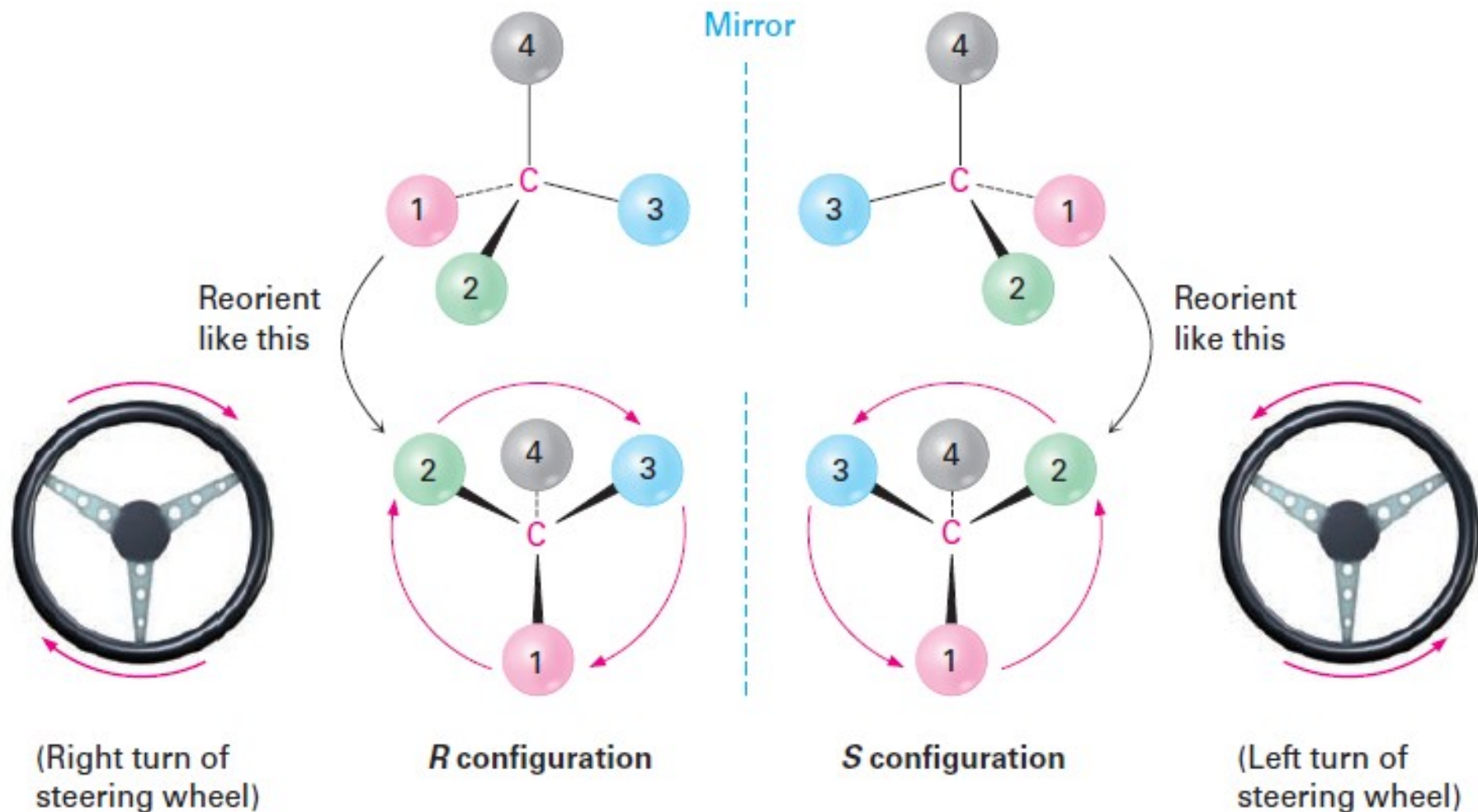


1.5. Naming enantiomers...

4. Mentally move from substituent priority 1 to 2 to 3.

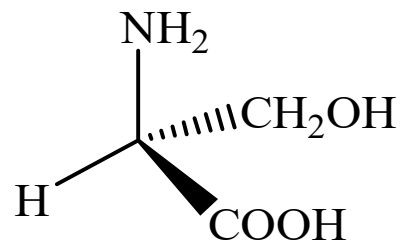


1.5. Naming enantiomers...

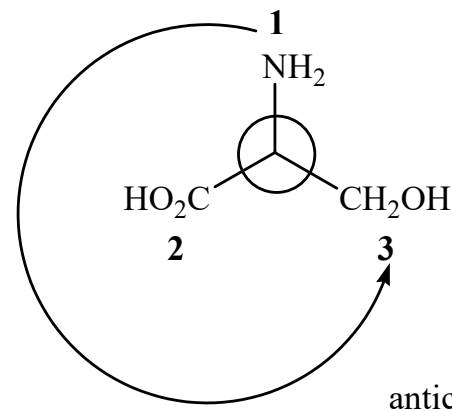
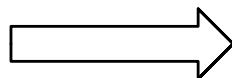
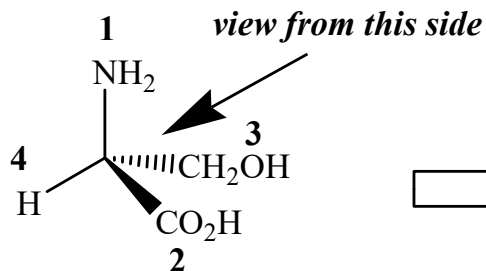


1.5. Naming enantiomers...

- Example: Serine



(-)-Serine



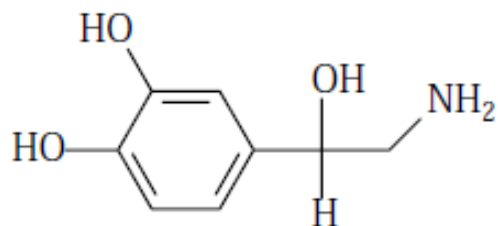
anticlockwise: *S*

Therefore, *S*-(*-*)-Serine

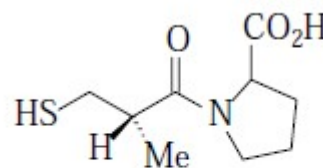
1.5. Naming enantiomers...

- **Exercise**

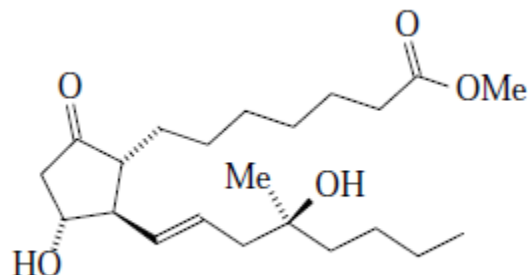
- Identify each chiral center as (R)- or (S)-



(-)-norepinephrine



captopril (antihypertensive)



misoprostol

1.6. Optical activity

1.6. Optical activity ...

- **Enantiomers**

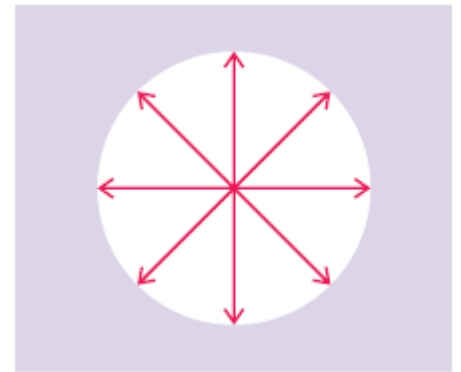
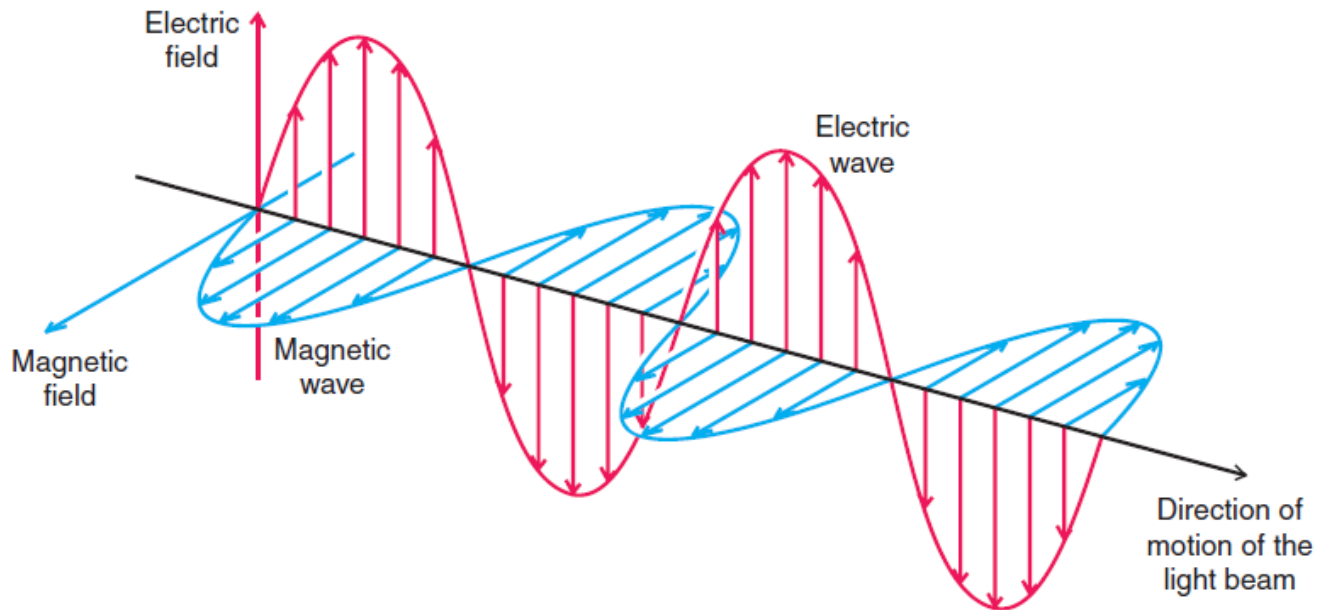
- Similar physicochemical properties

Compound	Boiling Point (bp) or Melting Point (mp)
(<i>R</i>)-2-Butanol	99.5°C (bp)
(<i>S</i>)-2-Butanol	99.5°C (bp)
(–)-(<i>R,R</i>)-Tartaric acid	168–170°C (mp)
(–)-(<i>S,S</i>)-Tartaric acid	168–170°C (mp)
(+/-)-Tartaric acid	210–212°C (mp)

- Except interaction with **plane polarized light**

1.6. Optical activity ...

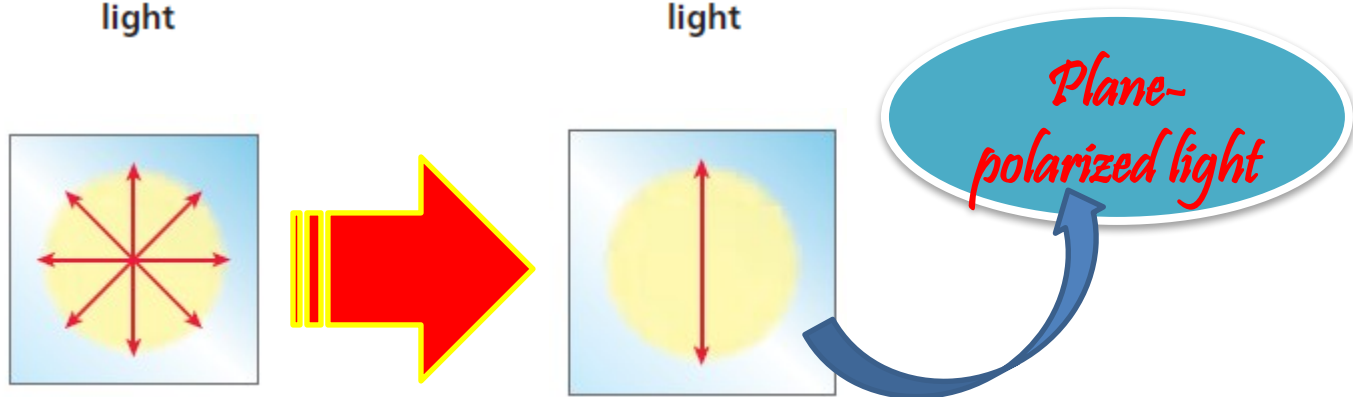
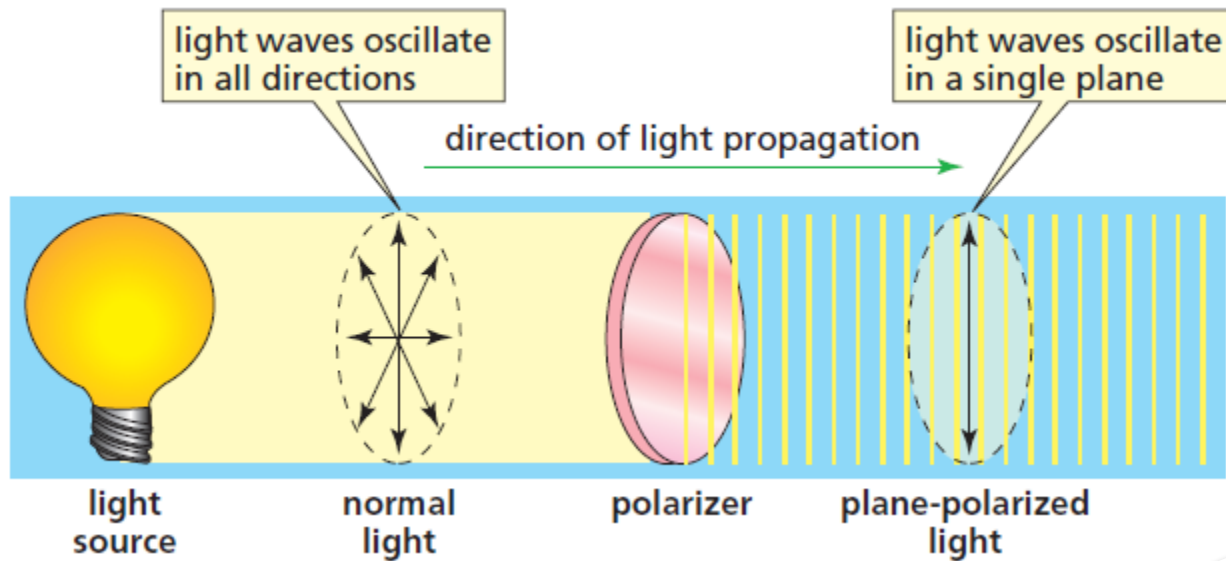
- **Light**



1.6. Optical activity ...

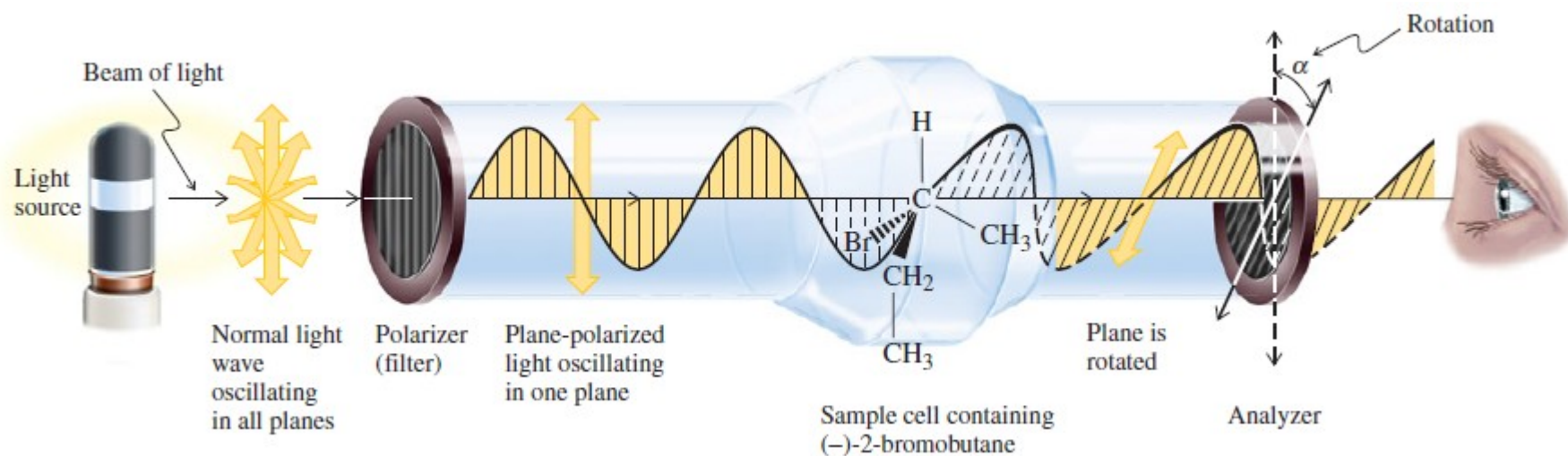
- **Polarizer**

- Interacts with the electric field component of light



1.6. Optical activity ...

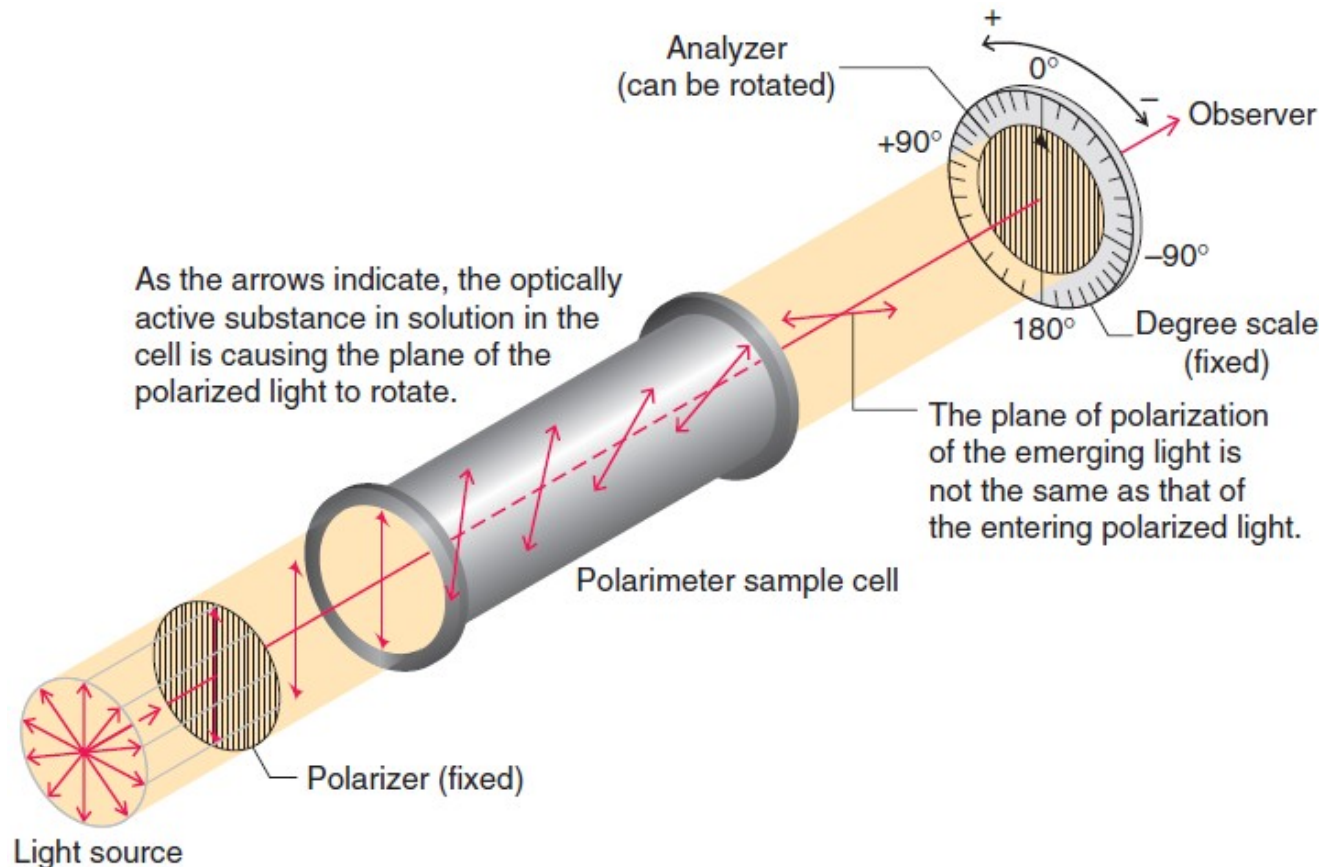
- **Polarimeter**
 - Optical rotation



1.6. Optical activity ...

– If rotation is

- Clockwise: (+); dextrorotatory
- Counterclockwise: (–); levorotatory



1.6. Optical activity ...

- **Observed rotation (α)**
 - Is the angle through which a sample of a compound (usually a solution) rotates plane-polarized light
 - Depends on:
 - *path length*
 - *concentration,*
 - *temperature,*
 - *solvent, and*
 - *wavelength*

1.6. Optical activity ...

- Specific for each optically active compound
- The specific rotation ($[\alpha]_{\lambda}^T$ (solvent))
“is the *number of degrees of rotation* caused by a solution of *1.0 g of the compound per mL* of solution in a *sample tube 1.0 dm* long at a *specified temperature and wavelength*”

1.6. Optical activity ...

- Hence, specific rotation ($[\alpha]_{\lambda}^T$ (solvent)) is defined as:

$$[\alpha]_{\lambda}^T(\text{solvent}) = \frac{\alpha}{l \times c}$$

- Example:
- When 0.300 g of sucrose is dissolved in 10.0 mL of water and placed in a sample cell 10.0 cm in length, the observed rotation is $+1.99^\circ$ (using the D line of sodium at 20°C). Calculate the specific rotation of sucrose.

1.6. Optical activity ...

- **Given:**

- $\alpha = +1.99^\circ$

- 0.300 g in 10 ml  $c = 0.300 \text{ g}/10.0 \text{ mL} = 0.03 \text{ g/mL}$

- $l = 10 \text{ cm} = 1 \text{ dm}$

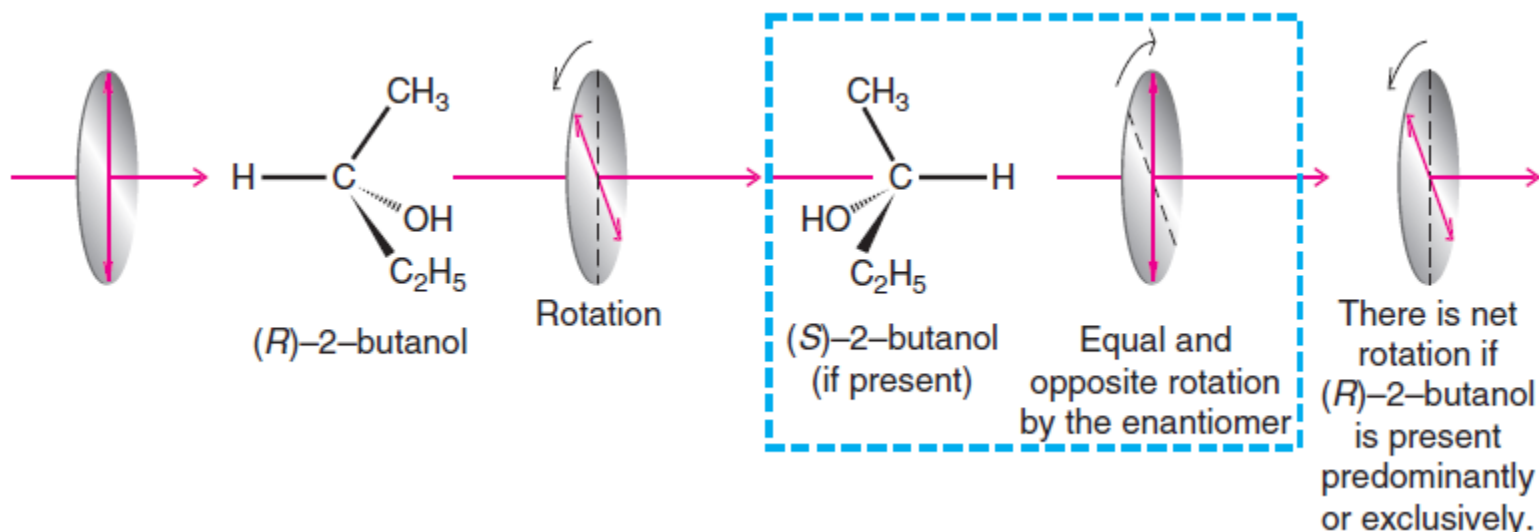
- **Required:** $[\alpha]_D^{20^\circ C}$

- **Solution:**

$$[\alpha]_D^{20^\circ C}(\text{water}) = \frac{\alpha}{l \times c} = \frac{+1.99^\circ}{1.00 \text{ dm} \times 0.03 \frac{\text{gm}}{\text{ml}}} = +66.3$$

1.6. Optical activity ...

- **Racemic mixtures**
 - *Mixtures containing equal quantities of enantiomers*
 - α is zero



1.6. Optical activity ...

- **Optical activity of a sample: ratio of enantiomers**
 - If only one enantiomer present
 - **Optically pure, or enantiomerically pure**
 - If equal
 - **Racemic**
 - One in excess
 - **Net rotation**
 - **Enantiomer excess (ee)**

1.6. Optical activity ...

- **Enantiomer excess (ee)**

- *tells us how much of excess of one enantiomer is in the mixture:*

$$\% \text{ Enantiomeric excess} = \frac{\text{moles of one enantiomer} - \text{moles of other enantiomer}}{\text{total moles of both enantiomers}} \times 100$$

- *Or*

$$\% \text{ Enantiomeric excess} = \frac{\text{observed specific rotation of the mixture}}{\text{specific rotation of the pure enantiomer}} \times 100$$

1.6. Optical activity ...

- **Example:**

- A solution of (+)-alanine from a fossil exhibits a value of $[\alpha] = 4.25$.
 - i. What is its *ee* and optical purity?
 - ii. What is the actual enantiomer composition of the sample, and how is the measured optical rotation derived from it?
- The specific rotation of pure (+)-alanine is + 8.5

1.6. Optical activity ...

– **Given:**

- $\alpha_o = + 4.25$
- $\alpha_s = + 8.5$

– **Required:** i. ee ii. Actual composition iii. $\alpha_o = 0.5 \alpha_s$

– **Solution**

i. % ee

$$\% \text{ Enantiomer excess (ee)} = \text{optical purity} = \frac{\alpha_o}{\alpha_s} \times 100 = \frac{4.25}{8.5} \times 100 = 50\%$$

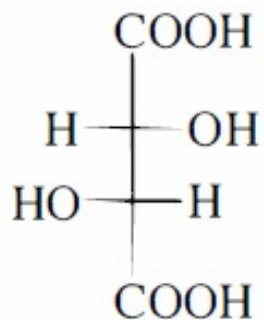
ii. 75% (+)- and 25% (–)-alanine

iii. $\alpha_o = 0.5 \alpha_s$

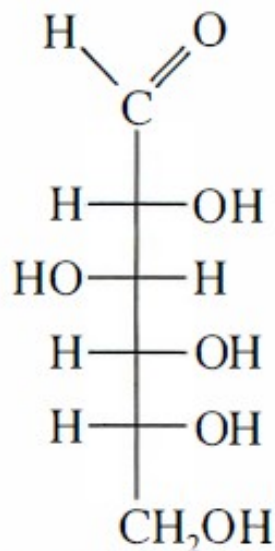
1.7. Resolution of enantiomers

1.7. Resolution...

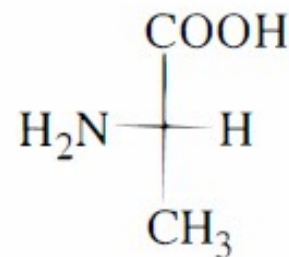
- **Nature:**
 - Pure enantiomers



L-(+)-tartaric acid
(2*R*,3*R*)-tartaric acid



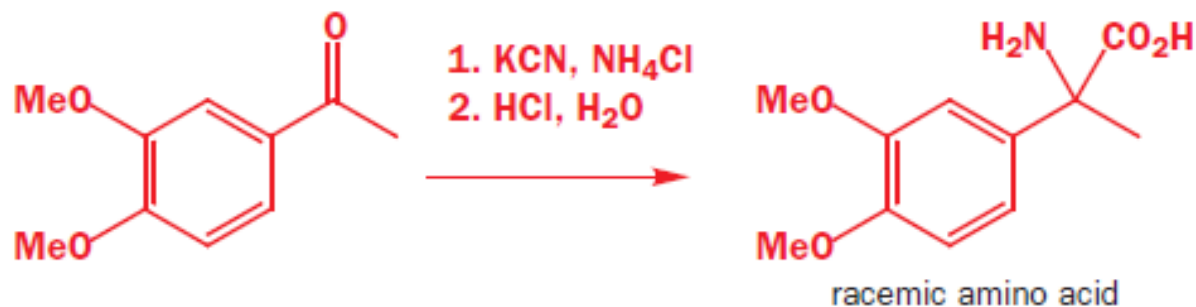
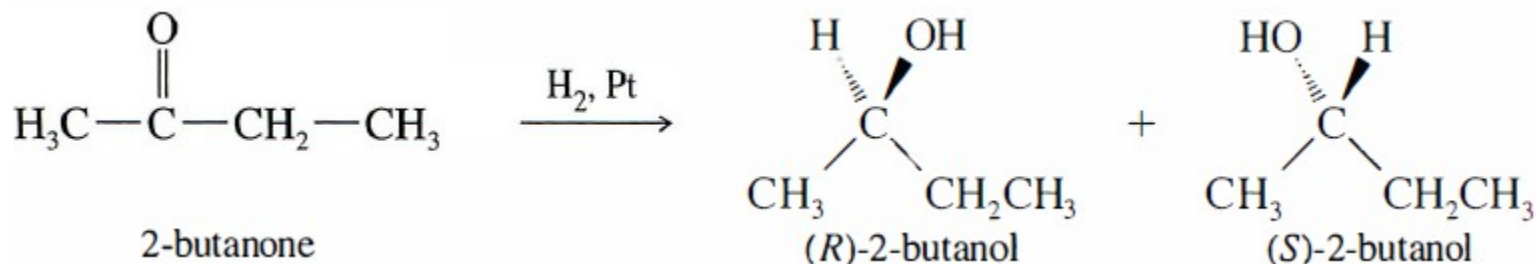
D-(+)-glucose



L-(+)-alanine

1.7. Resolution...

- Laboratory synthesis from achiral reagents
 - Racemate



1.7. Resolution...

- **Resolution**

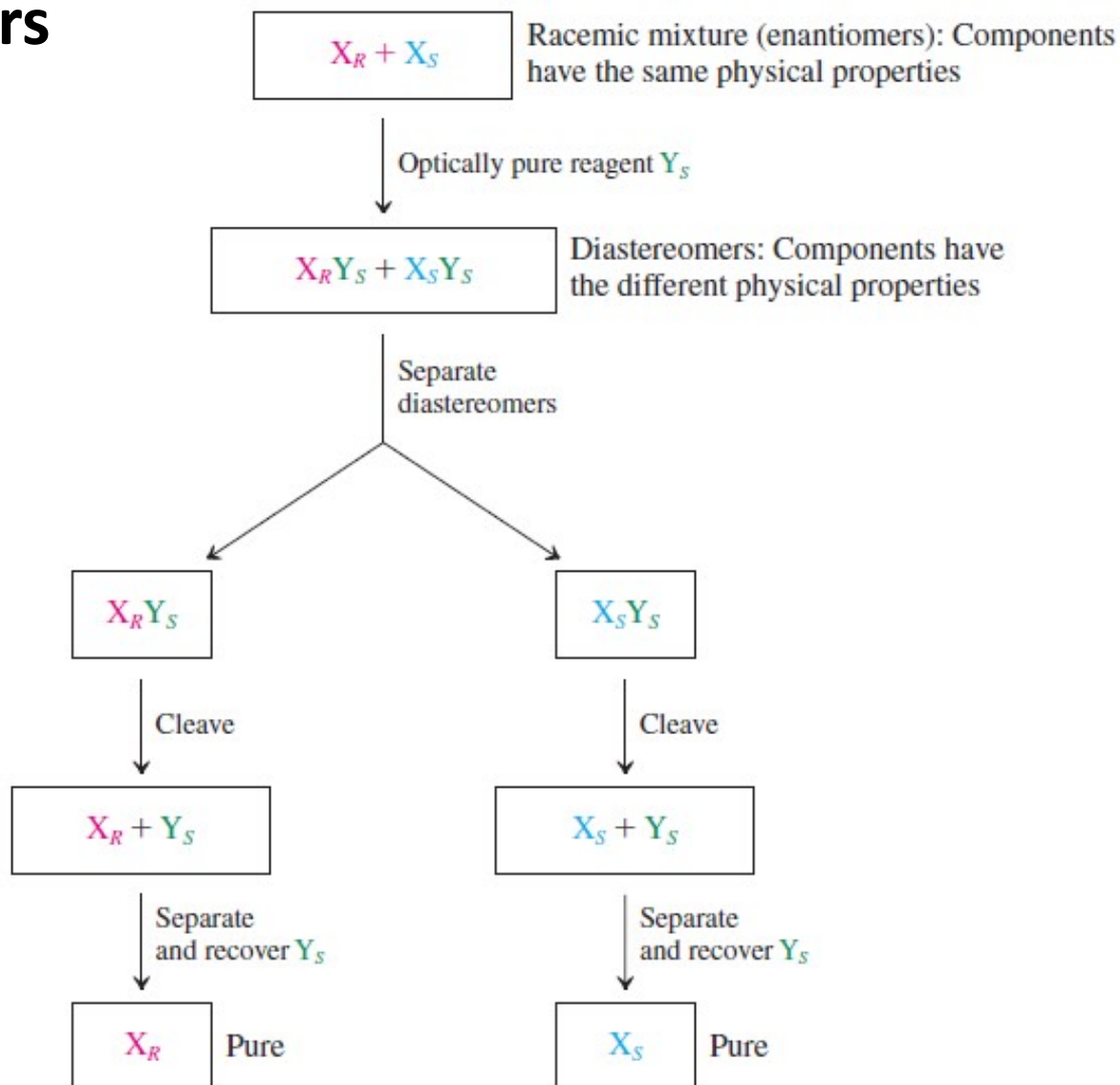
- *The separation of a racemic mixture into its enantiomeric components*

- How?

- Reaction with pure enantiomer
 - Enzymes
 - Chiral chromatography

1.7. Resolution...

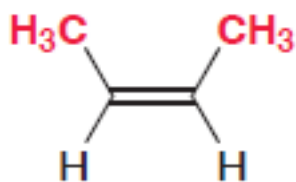
- Reaction with pure enantiomer: formation of diastereomers



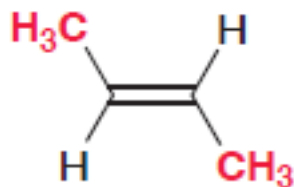
1.8. Diastereoisomers

1.8. Diastereoisomers ...

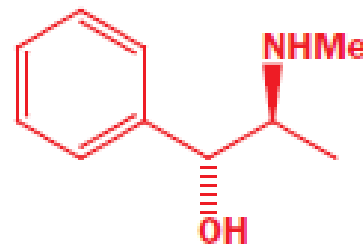
- **Diastereoisomers (diastereomers)**
 - Stereoisomers that are not mirror images of one another



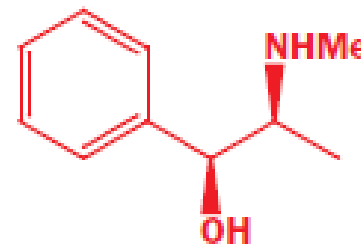
cis-2-Butene



trans-2-Butene



ephedrine

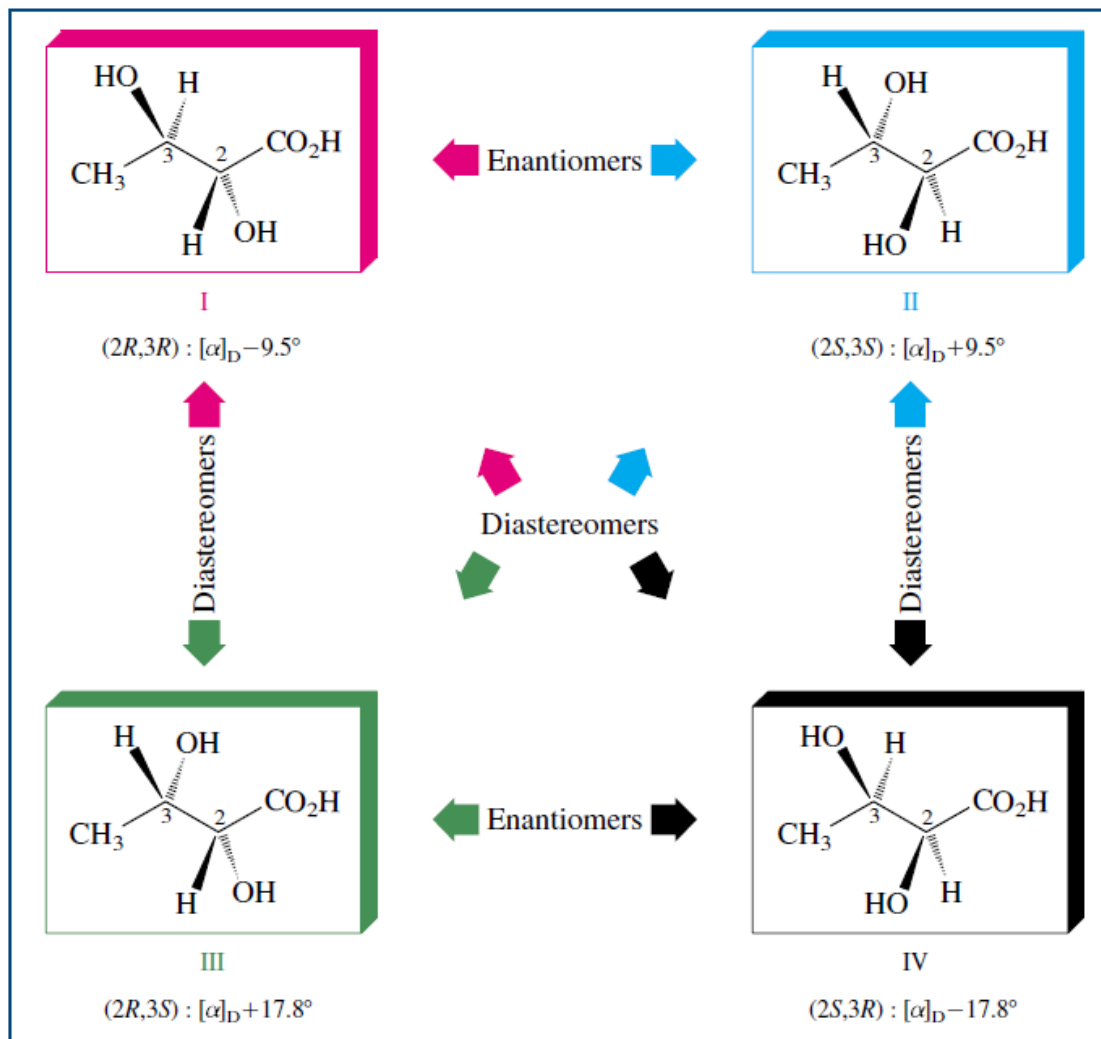


pseudoephedrine

- **Are either**
 - Geometric isomers
 - Compounds containing two or more chirality centers

1.8. Diastereoisomers ...

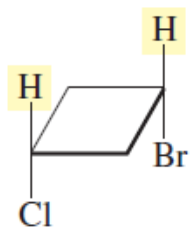
– Different physicochemical properties



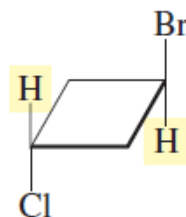
1.8.1. *Geometric isomers*

1.8.1. Geometric...

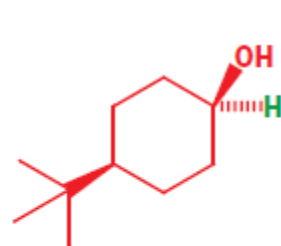
- **Geometric isomers (*cis-trans* isomers)**
 - Exist due to *hindred rotation about a bond*
 - Ring structure



cis-1-bromo-3-chlorocyclobutane



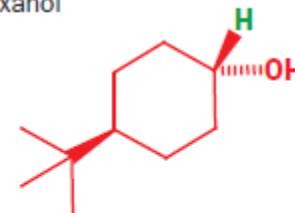
trans-1-bromo-3-chlorocyclobutane



cis isomer
cis 4-*t*-butylcyclohexanol
mp 82–83 °C

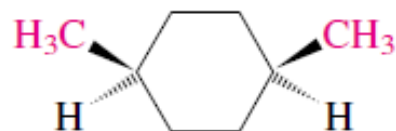
^1H NMR: δ_{H} of green proton 4.02

4-*t*-butylcyclohexanol

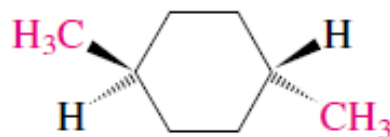


trans isomer
trans 4-*t*-butylcyclohexanol
mp 80–81 °C

^1H NMR: δ_{H} of green proton 3.50



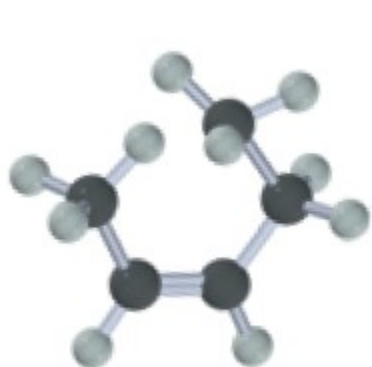
cis-1,4-Dimethylcyclohexane



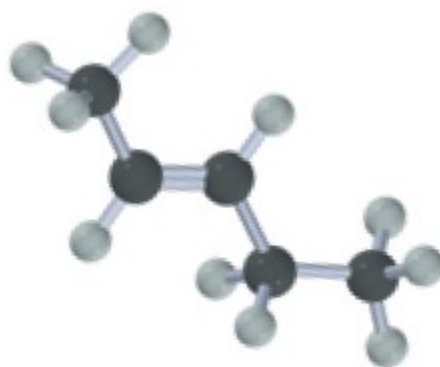
trans-1,4-Dimethylcyclohexane

1.8.1. Geometric...

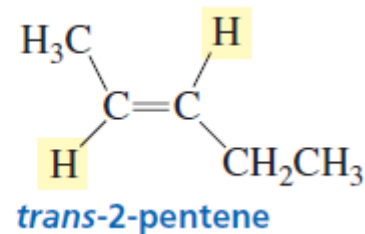
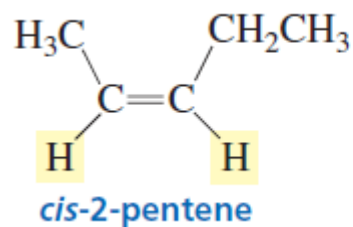
- Presence of multiple bonds



cis-2-pentene



trans-2-pentene



butenedioic acids



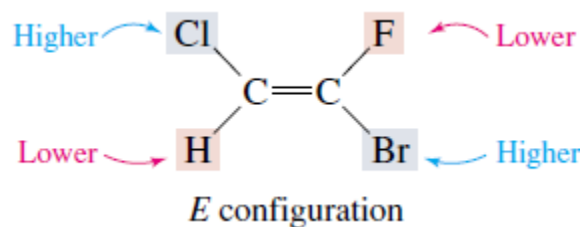
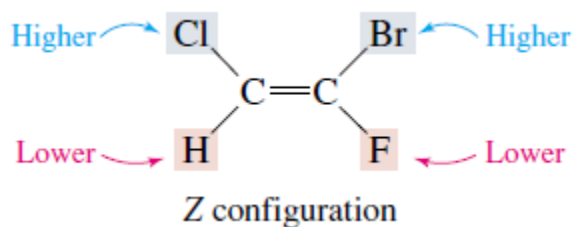
fumaric acid



maleic acid

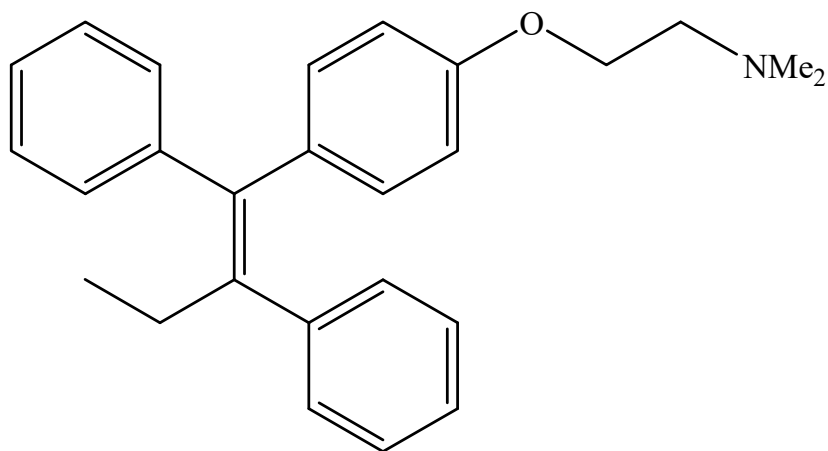
trans-butenedioic acid (fumaric acid)
m.p. 299–300 °C

cis-butenedioic acid (maleic acid)
m.p. 140–142 °C

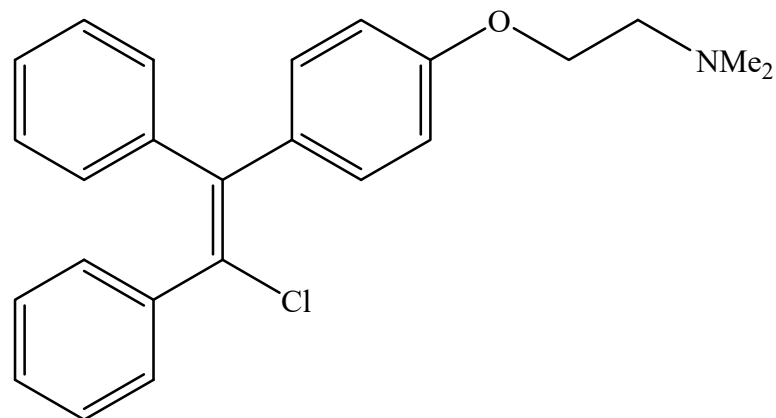


1.8.1. Geometric...

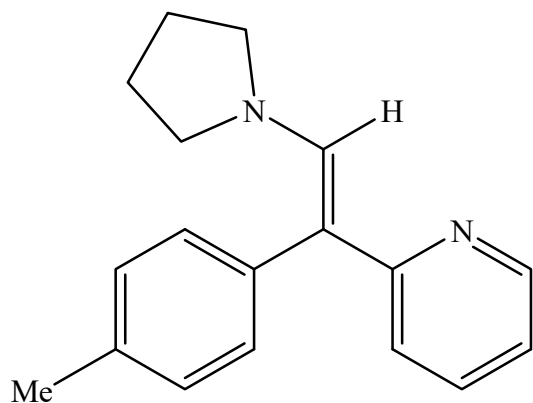
- Examples of drugs with geometric isomerism



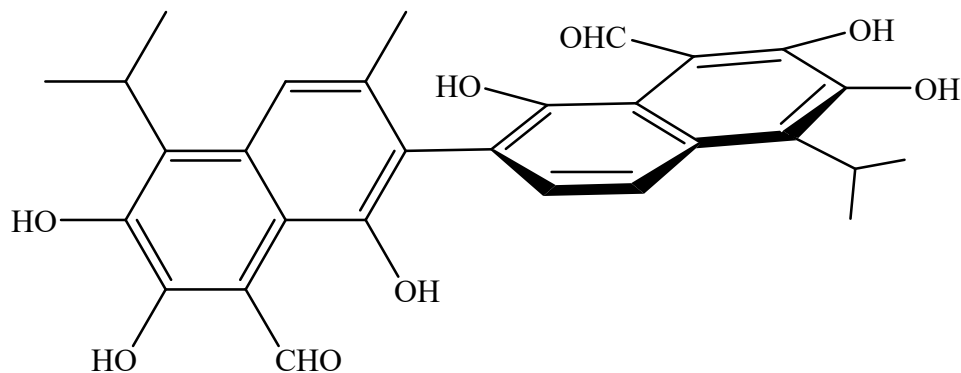
Tamoxifen



Clomifene



Triprolidine

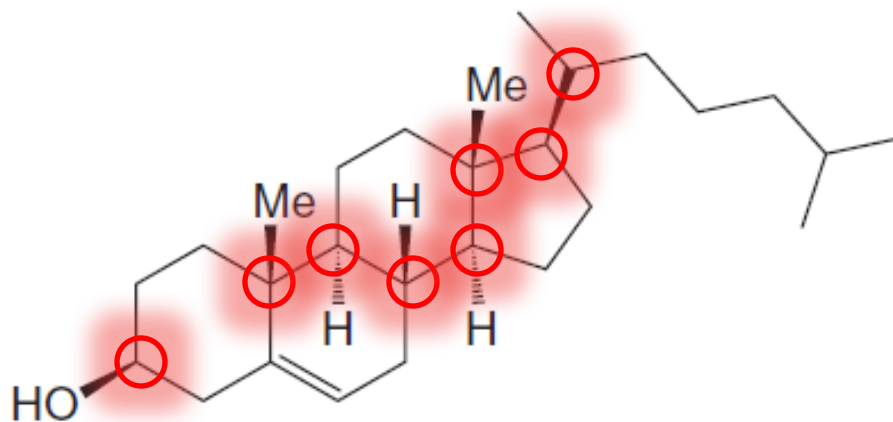


Gossypol

1.7.2. Isomers with more than one asymmetric carbon (chiral centers)

1.7.2. Isomers with...

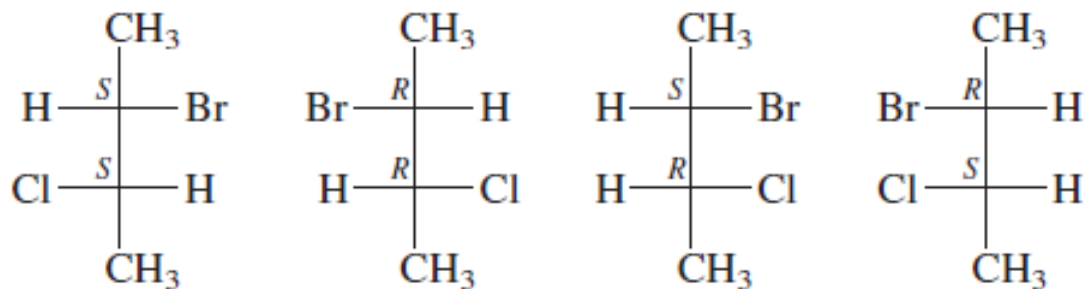
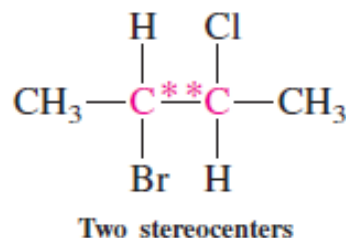
- **Molecules with many stereocenters**
 - How many stereocenters cholesterol has?



Cholesterol

1.7.2. Isomers with...

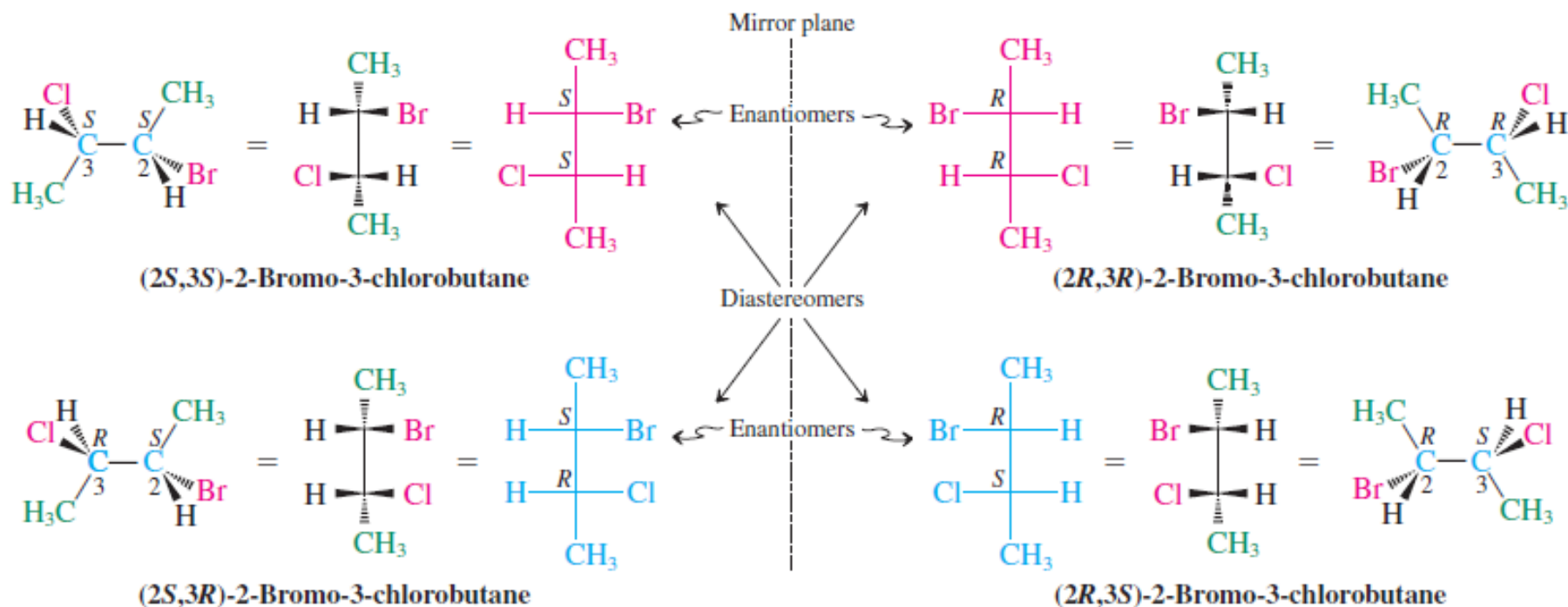
- For a molecule with n stereocenters
 2^n stereoisomers (not always. Why?)
- Example:
 - 2-Bromo-3-chlorobutane
 - 2 stereocenters
 - $2^2 = 4$ stereoisomers



Four stereoisomers of 2-bromo-3-chlorobutane

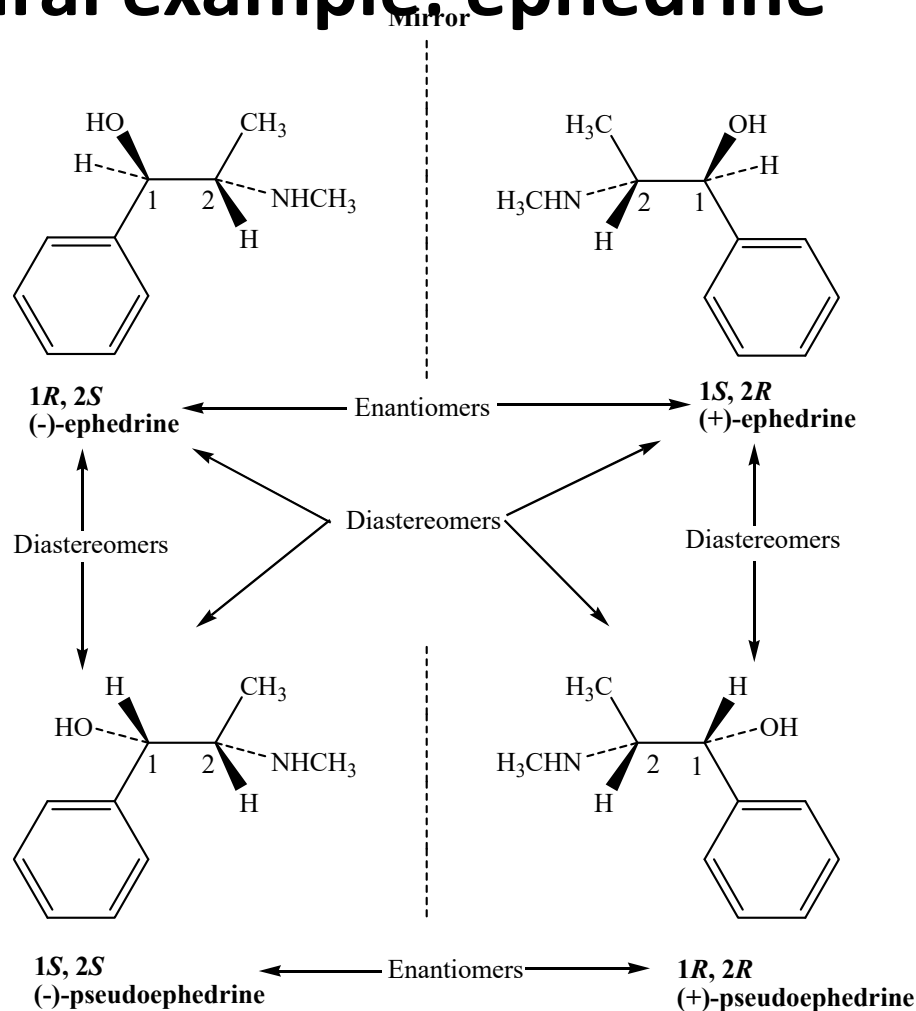
1.7.2. Isomers

Another example: 2-Bromo-3-chlorobutane



1.7.2. Isomers with...

- One more natural example: ephedrine

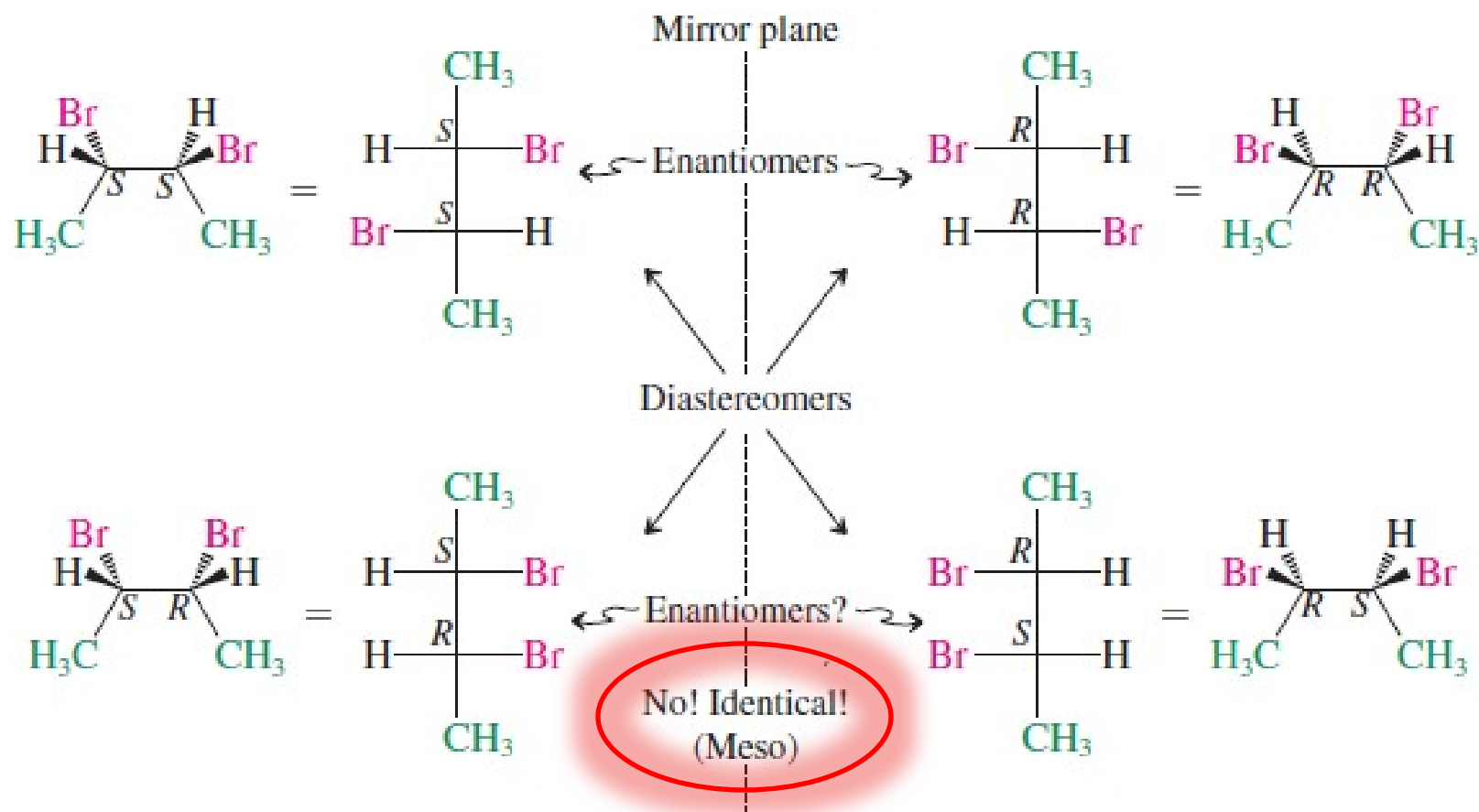


	(1R,2S)-(-)-ephedrine	(1S,2R)-(+)-ephedrine	(1S,2S)-(+)-pseudoephedrine	(1R,2R)-(-)-pseudoephedrine
m.p.	40–40.5 °C	40–40.5 °C	117–118 °C	117–118 °C
$[\alpha]_D^{20}$	-6.3	+6.3	+52	-52

1.8. *Meso compounds*

1.8.

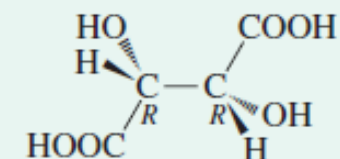
Meso...



1.8. Meso...

- **Meso compounds**

- Compounds that are achiral, yet contain chirality centers
- How to check?
 - Plane of symmetry?
 - Best example: tartaric acid
 - two stereocenters; 4 stereoisomers ????

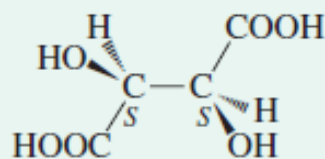


(+)-Tartaric acid

$$[\alpha]_D^{20^\circ\text{C}} = +12.0$$

m.p. 168–170°C

Density (g mL⁻¹) $d = 1.7598$

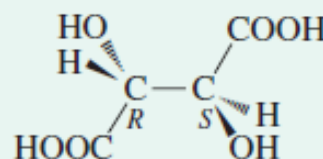


(-)-Tartaric acid

$$[\alpha]_D^{20^\circ\text{C}} = -12.0$$

m.p. 168–170°C

$d = 1.7598$



meso-Tartaric acid

$$[\alpha]_D^{20^\circ\text{C}} = 0$$

m.p. 146–148°C

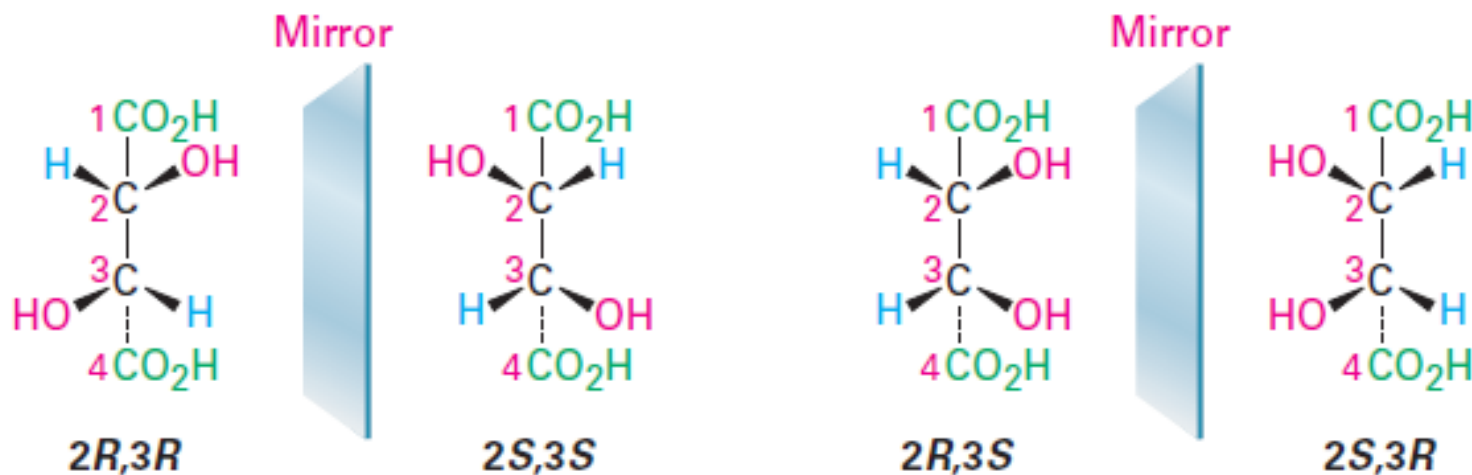
$d = 1.666$



Pasteur's polarimeter and crystals of (+)- and (-)-tartaric acid.

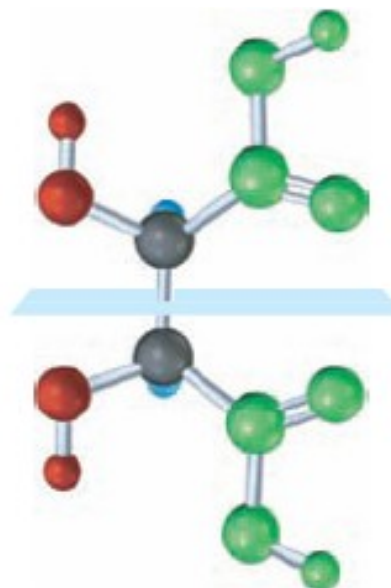
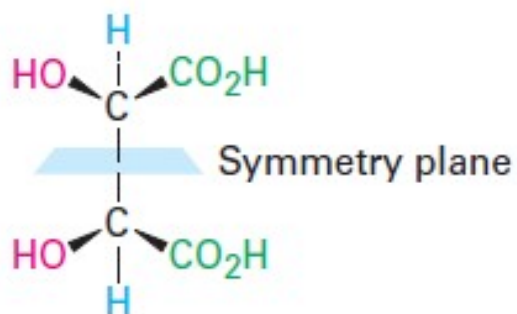
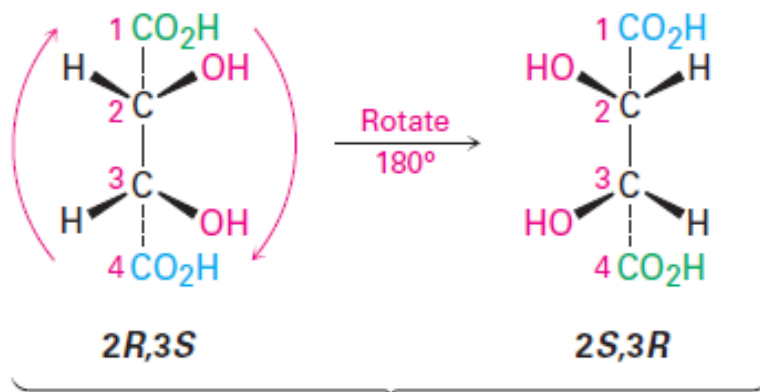
1.8. Meso...

- Why only three?



1.8. Meso...

- But the 2R,3S and 2S, 3R really enantiomers?
 - No!!!!



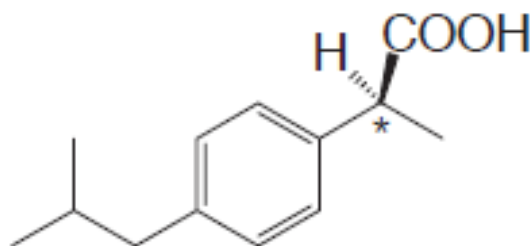
1.10. Biological significance of stereoisomerism

1.10. Biological...

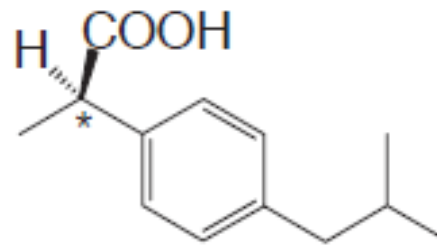
- **Drugs as well as other biological compounds**
 - Show stereoisomerism
 - One enantiomeric form of a drug may be active, and the other may be
 - inactive,
 - less active or
 - even toxic.

1.10. Biological...

- Examples:
 - Ibuprofen



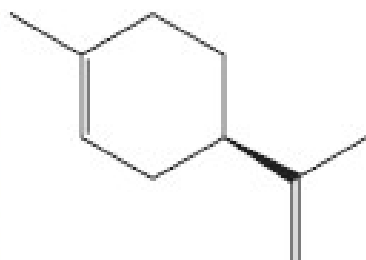
(*S*)-Ibuprofen
Active stereoisomer



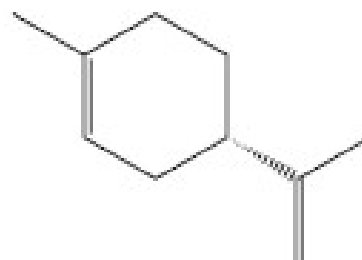
(*R*)-Ibuprofen
Inactive stereoisomer

1.10. Biological...

– Limonene



(+)-Limonene
(the enantiomer
of limonene found
in oranges)



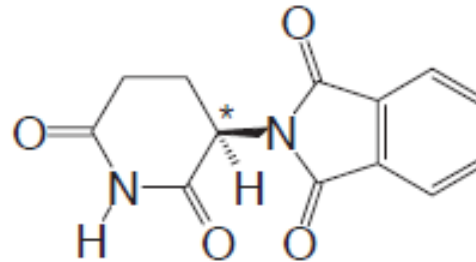
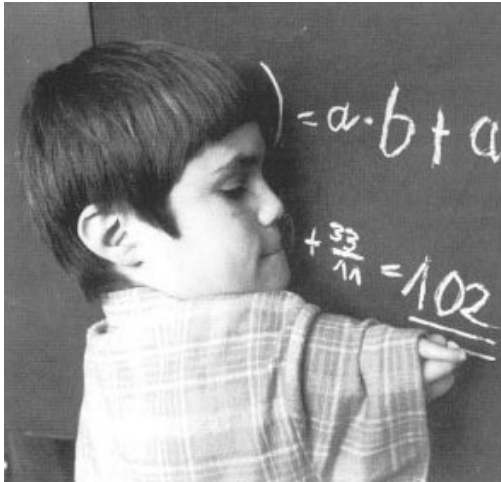
(-)-Limonene
(the enantiomer
of limonene found
in lemons)



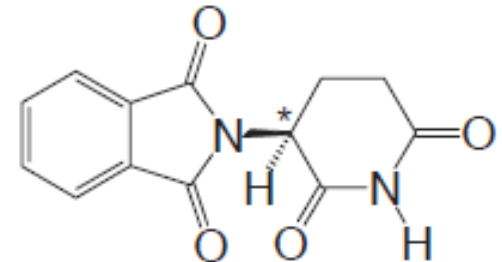
1.10. Biological...

– Thalidomide

- The Thalidomide Tragedy: 1961



Sedative



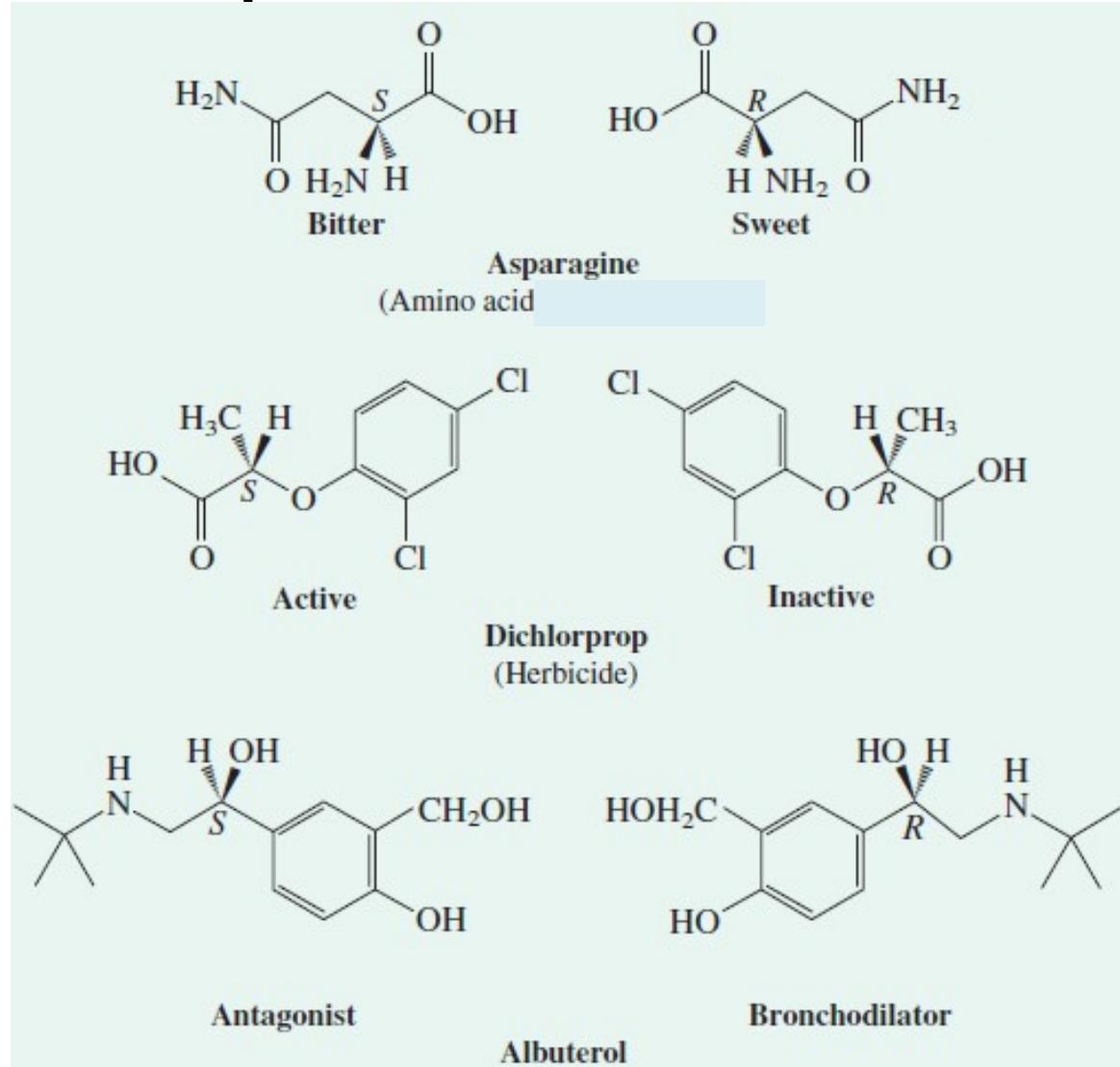
Teratogenic

Thalidomide stereoisomers



1.10. Biological...

– Further examples



1.10. Biological...

- ***Why do enantiomers have different biological properties?***
 - Most of the molecules that make up living organisms are chiral

